Hands-On Data Analysis Using Jamovi

Volume I: Foundations and Core Statistical Methods for Health Researchers

Dr Naveen Suthar

5 October 2026

Review manuscript, revision 2

The teaching text and conceptual exercises have been prepared for author review. The Hands-on note boxes are intentionally empty and reserved for the author's jamovi datasets, saved analyses, screenshots, and procedures.

All numerical health examples are invented for teaching. They do not represent observed patient data or clinical recommendations.

Copyright © 2026 Dr Naveen Suthar. All rights reserved.

Contents

Preface 1

I Foundations 4

1 Role of Statistics in Health Research 6

2 Getting Started with jamovi 11

3 Data Entry and Validation 15

4 Data Cleaning and Missing Values 20

5 Transformations and Derived Variables 25

6 Preparing Data for a Specific Analysis 30

II Descriptive Statistics and Graphs 36

7 Descriptive Statistics 38

8 Graphical Presentation 43

III Core Inferential Methods 48

9 Estimation and Hypothesis Testing 50

10 Comparing Two Groups or Two Paired Measurements 56

11 Comparing More Than Two Groups 62

12 Associations Between Categorical Variables 67

IV Correlation and Introductory Regression 72

13 Correlation 74

14 Introduction to Simple Linear Regression 79

15 Introduction to Logistic Regression 84

V Reporting and Interpretation **89**

16 Reporting Statistical Results **91**

Appendices **96**

A Statistical Formulas **96**

B Choosing an Appropriate Statistical Method **101**

C Glossary **105**

D Practice Questions and Applied Cases **109**

References **115**

Preface

Statistical analysis begins with a research question and ends with a statement that another reader can evaluate. Between these points, the researcher must define the population, understand how observations were collected, choose an appropriate summary or model, examine assumptions, and explain uncertainty. Software supports these decisions; it does not make them on the researcher's behalf.

This book is written for medical students, clinicians, postgraduate trainees, and health researchers beginning applied data analysis. Its purpose is to connect a clinical question with the structure of the data, the statistical method, and the meaning of the result. The emphasis is on estimates and their uncertainty, supported by appropriate tests and clear reporting. The American Statistical Association's statement provides the basis for interpreting a p-value in context rather than treating a threshold as a complete scientific conclusion. [1]

The author dedicates this work to Lord Jagannath. The project began on Ram Navami, 26 March 2026. This version is review revision 2, dated 5 October 2026.

What this volume covers

Volume I covers the foundations needed to analyse a straightforward health dataset: study design, variable types, data entry and validation, cleaning and missing values, transformations, preparation for a specific analysis, descriptive statistics, graphs, uncertainty, group comparisons, categorical associations, correlation, simple linear regression, and reporting. These topics are developed in 16 teaching chapters. An introductory logistic regression chapter explains binary outcomes and odds ratios as a bridge to Volume II. Adjusted modeling, advanced diagnostic assessment, and prediction-model development belong to later volumes.

After studying this volume, the reader should be able to define the unit of observation; distinguish independent from paired observations; check data coding and missingness; choose a basic method for a specified question; interpret an estimate with its confidence interval; and write a results paragraph supported by a clear table or figure. These are the learning outcomes used throughout the manuscript.

How to use the book

Read the foundations before moving to an analysis chapter. Within each chapter, start with the question and data structure, then study the method, assumptions, illustrative example, interpretation, and reporting. Complete the review questions before checking the answers. The decision guide in the appendices supports method selection but does not replace reading the relevant chapter.

Every main chapter contains a separate **Hands-on** subsection and a note box. These boxes are intentionally empty in this review version. They are reserved for the author's jamovi datasets, screenshots, saved analyses, and procedural demonstrations. The conceptual text and small

numerical illustrations can be reviewed now; the complete practical workflow will be added in those boxes.

All health examples and numerical results in this manuscript are invented for teaching. They do not represent observed patient data, claims about treatment effectiveness, or clinical thresholds. A number used to illustrate a method should not be taken as a recommendation for patient care.

The three-volume progression

Volume	Central task	Boundary
I: Foundations and core methods	Describe, compare, associate, and report	Basic independent or paired analyses; simple regression; introductory binary-outcome interpretation
II: Adjusted models and specialist methods	Adjust, model, diagnose, and interpret	Multiple linear and logistic regression; count and risk models; survival; diagnostic tests; measurement; repeated measures
III: Advanced inference and reproducible analysis	Assess validity, validate, synthesise, and reproduce	Causal thinking; advanced models; prediction; evidence synthesis; detailed jamovi–R workflows

The recurring sequence is **question, design, data, method, assumptions, output, clinical meaning, and reporting**. Reproducibility begins in Volume I through data documentation and saved analysis files. Later volumes extend that foundation to more complex modeling and code-based workflows.

Reading statistical language

An **outcome** is the variable whose distribution or relationship is being studied. A **predictor** is a variable used to explain or model the outcome. These labels describe an analysis role and do not establish a causal relationship. A **parameter** is a population quantity; a **statistic** is calculated from the observed sample. An **estimate** is the sample-based value used to learn about a parameter.

Read a result in three parts: its direction and size, its uncertainty, and its relevance to the research question. For example, a mean difference is incomplete without the group order and measurement unit. A ratio is incomplete without its reference category. A p-value is incomplete without the model and hypothesis to which it refers. Reporting guidance is available in the SAMPL guidelines. [2]

About the author

Dr Naveen Suthar is the author of this series. Project correspondence can be directed to books.dr naveen@gmail.com. The manuscript is maintained at https://github.com/DrNaveen01/Jamovi-book-1.

Abbreviations

Abbreviation	Meaning
ANOVA	Analysis of variance
CI	Confidence interval
ID	Identifier
IQR	Interquartile range
OR	Odds ratio
Q–Q	Quantile–quantile
RD	Risk difference, when the design supports a risk interpretation
RR	Risk ratio, when the design supports a risk interpretation
SD	Standard deviation
SE	Standard error

Copyright

Copyright © 2026 Dr Naveen Suthar. All rights reserved. Third-party software, documentation, trademarks, and later-added datasets or screenshots retain their respective ownership and licence terms. The bibliography identifies sources supporting the statistical and software guidance.

Part I

Foundations

The first task is to establish what the data can answer. A well-defined question identifies the population, the outcome, the comparison or predictor, and the time at which the outcome is measured. The study design then determines whether observations are independent, paired, repeated, or clustered. These decisions must precede a statistical test.

The six chapters in this part move from the research question to a usable analytical record. The first introduces designs, measurement scales, sampling, and the unit of observation. The second explains the jamovi environment. Four dedicated preparation chapters then address data entry and validation, cleaning and missing values, transformations and derived variables, and preparation for a specific analysis.

These stages have different purposes. Validation discovers discrepancies between the file and the measurement plan. Cleaning resolves them using evidence. Transformation creates quantities needed for the question. Analysis preparation defines which eligible observations and variables contribute to a particular result. Keeping those purposes explicit helps the researcher explain both the values and the denominator.

Use this part to prepare an analysis specification containing the question, eligible observations, variables, coding, missing-value rules, and intended summaries. STROBE reporting guidance asks observational researchers to describe participants, variables, bias, study size, and statistical methods explicitly. That reporting requirement is easier to meet when these items are documented before analysis. [3]

By the end of this part, the reader should be able to inspect a dataset without starting a test and identify whether its structure matches the intended analysis.

Chapter 1

Role of Statistics in Health Research

1.1 Learning objectives

After this chapter, the reader should be able to translate a health question into an analysis objective; identify the population and unit of observation; distinguish common variable types; recognise independent and paired data; and separate sampling uncertainty from problems in study conduct.

1.2 Start with the question

Consider three questions about an outpatient clinic: What proportion of attendees have elevated blood pressure? Does mean blood pressure differ between two treatment groups? Does blood pressure change within a patient after follow-up? These questions require different data structures and target quantities, even though all involve the same measurement.

Write the question as a sentence before naming a test. Specify who is included, what is measured, which comparison is relevant, and when the measurement occurs. For example: "Among eligible adults attending the clinic during the study period, what is the mean difference in systolic blood pressure between group A and group B at three months?" This sentence defines the population, outcome, groups, time point, and direction of the intended comparison.

The **analysis objective** describes what the analysis should do. Description summarises observed values. Estimation quantifies a population quantity. Comparison estimates a contrast between groups. Association examines how variables vary together. Prediction estimates an outcome for an individual or future observation. A study can contain more than one objective, but each analysis should have a defined role.

1.3 Define the population, sample, and target quantity

The population is the set of people or observations to which the question refers. The sample is the set actually observed. A clinic sample and a community population are different targets: attendance may depend on illness, access, referral, or willingness to seek care. State the population to which the findings are intended to apply, rather than allowing the software's sample size to define it.

A parameter describes a population quantity, such as a mean or proportion. A statistic is its sample-based counterpart. The **estimand** is the precise quantity the study seeks to estimate. "Difference in blood pressure" is incomplete; "mean follow-up blood pressure in A minus that in B among eligible adults" is more precise. A positive or negative value then has a defined meaning.

Record the scale of the estimand. A difference uses the original measurement unit. A proportion describes a fraction of a defined denominator. A ratio compares two quantities on a multiplicative scale. Each requires a different reporting sentence, even when the underlying research concern is similar.

1.4 Understand the study design

Design	What is observed	Basic analytical implication
Cross-sectional study	Variables measured within a specified period	Describe prevalence and concurrent associations; clarify the sampling frame
Cohort study	Outcomes observed over follow-up in defined groups	Specify eligibility, follow-up, and event ascertainment before interpreting risk
Case-control study	Participants selected according to outcome status	The sampled case fraction is not the population risk; an odds ratio may be appropriate
Randomised trial	Intervention assignment is randomised	Preserve the planned comparison and account for the randomisation unit
Before–after study	The same individuals measured at two times	Measurements are paired; a change estimate requires complete pairs or another specified method
Clustered study	People grouped within facilities, households, or other clusters	Standard independent-observation methods may need extension

This table is an orientation to data structure, not a substitute for a design-specific analysis plan. STROBE identifies the observational designs covered by its statement and the information needed to describe their conduct and analysis. [3]

Random assignment and random sampling answer different concerns. Assignment determines how study groups are formed; sampling determines how observations enter the study. A trial can randomise treatment among a selected clinic population without representing every person in the wider community. Keep the study's internal comparison and its intended generalisation as separate judgments.

1.5 Identify the unit of observation

The unit of observation is the entity represented by one analytical record. It may be a patient, hospital admission, visit, household, or facility. It is not automatically a person. If a patient has several admissions, counting each admission creates a study of admissions unless the analysis explicitly accounts for repeated records from the same person.

For a simple independent analysis in this volume, one row usually represents one participant and each participant contributes once to the relevant comparison. For a paired analysis, the same participant contributes measurements at two times or a defined matched pair contributes two related observations. More extensive repeated or clustered structures require methods developed in later volumes.

Before analysis, complete the sentence: "One row represents …" Then count unique units as well as rows. An accidental duplicate row and a valid repeat visit can look similar in a spreadsheet, but they have different meanings and should not be handled by the same deletion rule.

1.6 Types of data and measurement scales

Variable type	Meaning	Illustrative health variable	Suitable initial summary
Nominal	Categories without an intrinsic order	Blood group	Counts and proportions
Binary	A nominal variable with two levels	Event present or absent	Event count and denominator
Ordinal	Ordered categories without a guaranteed equal spacing	Mild, moderate, severe	Ordered counts and proportions
Continuous	Quantitative measurement on a meaningful numerical scale	Systolic blood pressure in mmHg	Distribution, central value, and spread
Count	Number of occurrences	Number of admissions	Distribution, zeros, range, and an appropriate count summary
Identifier	A label linking records to a unit	Study participant code	Uniqueness and linkage checks

In jamovi, measure type and storage type are separate properties. An integer column can contain a continuous measurement, an ordered score, a category code, or an identifier. Setting a column as nominal, ordinal, continuous, or ID determines how it is treated in analyses; a numerical code alone does not establish its measurement scale. [4]

Age recorded in years is quantitative even when stored as whole numbers. A code of 1 for a blood group is a label, not a quantity to average. A severity category coded 1, 2, and 3 is ordered, but the numerical distance between categories need not represent equal changes in severity. Preserve that distinction when choosing a method.

The role of a variable can change across questions. Age may be the outcome in a descriptive analysis and a predictor in a regression. Its measurement type remains the same; its analytical role changes. Define both properties in the data dictionary.

1.6.1 Interval and ratio measurement scales

An interval scale has meaningful equal differences but an arbitrary zero. Celsius temperature is a familiar example: a ten-degree difference has meaning, but 20°C is not twice the temperature of 10°C. A ratio scale also has a meaningful zero, allowing ratios such as one measured length being twice another. The distinction concerns the measurement definition, not how many decimal places are stored. [5]

Nominal, ordinal, interval, and ratio are the classical scale categories. Quantitative data can also be discrete, such as a count, or treated as continuous, such as a length measurement. These classifications answer different questions: ordering and meaningful arithmetic on one hand, and the set of possible numerical values on the other.

1.7 Sampling, uncertainty, and bias

Sampling variability means that repeated samples could give different estimates. A standard error or confidence interval addresses this variation under stated assumptions. A systematic measurement error, selective inclusion, or unaddressed confounding is a different problem. A narrow interval around a biased estimate remains misleading.

Use a simple thought experiment. Suppose a weighing scale adds two kilograms to every reading. Collecting many measurements can make the sample mean very precise, while the systematic two-kilogram error remains. The remedy is to check the measurement process; merely increasing the sample size does not correct the instrument.

Similarly, a large clinic dataset may describe clinic attendees well while failing to represent people who never attend. In the report, distinguish the uncertainty quantified by the statistical method from limitations in how the observations were obtained. Confidence intervals are conditional on the analysis assumptions and do not automatically include every source of uncertainty. [6]

1.8 Establish an analysis workflow

Start by specifying the question and design. Create a data dictionary and preserve the source file. Check records, coding, units, missingness, and eligibility. Describe the observed data and inspect relevant graphs. Select the estimand and method, examine assumptions, then interpret the estimate and its uncertainty. Finally, write the result and retain the analysis file and decision log.

The workflow should leave an audit trail. Record a change because it is justified by a coding definition or verified source, not because it improves a p-value. Record an exclusion because it follows the study rules, not because an observation is inconvenient. A documented decision can later be examined and revised; an undocumented change is difficult to reconstruct.

1.9 Illustrative analysis specification

The following specification is invented for teaching:

Item	Specification
Question	Compare mean systolic blood pressure at three months between A and B
Population	Eligible adults included in the study during its recruitment period
Unit	One participant
Outcome	Follow-up systolic blood pressure, mmHg
Groups	A and B; each participant belongs to one group
Contrast	A minus B
Main result	Mean difference with a 95% confidence interval
Checks	Group membership, outcome availability, independence, distribution, and unusual observations
Interpretation boundary	The strength of any causal claim depends on the design and conduct

This specification guides the later analysis. It makes clear why the chapter on two independent groups is relevant and why a paired test would answer a different question.

1.10 Hands-on

jamovi workspace

1.11 Review questions

1. A dataset contains 300 visits from 120 patients. What must be defined before calling its sample size 300?
2. A severity variable is coded 1, 2, and 3. What information is needed before treating it as continuous?
3. A confidence interval becomes narrower after increasing sample size. Which study problems can still remain?

1.12 Answers and reasoning

1. Define the analytical unit and whether repeated visits from the same patient are relevant and dependent. A study of patients and a study of visits have different denominators.
2. Establish what the categories mean and whether the scale has a justified quantitative interpretation. Numerical codes do not establish equal spacing.
3. Selection, measurement, and other systematic problems may remain. Greater precision does not establish representativeness or correct a biased measurement process.

Chapter 2

Getting Started with jamovi

2.1 Learning objectives

After this chapter, the reader should be able to distinguish the data, analysis, and output components of jamovi; inspect imported variable properties; recognise how file formats affect reproducibility; and prepare a saved analysis file for later review.

2.2 The purpose of the software

jamovi provides a spreadsheet-style data view and an analysis interface with results displayed alongside the data. Its official introductory guidance describes importing data, setting variable properties, running an analysis, and saving the data and analysis together in an `.omv` file. [7]

Treat the software as an environment for implementing an analysis plan. It can calculate results once variables and options are assigned, but it cannot determine whether the study question, population, eligibility rules, or causal interpretation are appropriate. A result appearing in the output panel confirms that a calculation was performed; it does not by itself validate the analysis.

2.3 Installation and version information

Obtain jamovi from the official website, https://www.jamovi.org. The author's Hands-on demonstration should identify the operating system, jamovi version, and any additional module used. The same analysis name can be available through different modules, and software updates can alter defaults or output labels.

Record versions at the beginning of a project and when an analysis is repeated after an update. If two researchers obtain different results, compare the data, variable settings, selected options, missing-data handling, and versions before assuming that one numerical result is wrong. Reproducibility requires matching these choices.

The learning aim is to understand the meaning of an option before selecting it. For example, selecting a confidence interval does not change the research question; it requests an uncertainty summary. Changing the reference category changes the direction of a reported contrast. These options have different analytical consequences.

2.4 Understand the interface

The data area contains observations and variables. The variable editor defines properties such as the name, data type, measure type, and levels. The analysis area assigns variables and options to a method. The results area displays tables and other output. The spreadsheet documentation explains the distinction between data variables, computed variables, and transformed variables. [8]

Use the data view to verify what will enter an analysis. Use the analysis options to verify what calculation is requested. Use the output to inspect the resulting estimate, sample size, uncertainty, and assumptions. Each area answers a different question, so a complete review should examine all three.

An automatically updated result can be useful during exploration, but it also means that a later change to the data or a filter can change previously generated output. Save dated versions at important review points and record intentional changes. Do not treat a screenshot as a substitute for the file that generated it.

2.5 Importing and structuring data

Prepare the source file with one header row, one variable per column, and one record per defined observation unit. Avoid merged cells, embedded totals, decorative headings, and values that combine a number with a unit in the same cell. A column of values such as "120 mmHg" requires a clear conversion to numeric values before being treated as a continuous measurement.

Maintain a source copy before changing the imported data. Compare the imported number of rows and variables with the expected source dimensions. Inspect several records at the beginning and end of the file, as well as unusual values, rather than assuming that a successful import preserved the intended meaning of every column.

Use short, stable variable names with descriptions for their full meaning. For example, `sbp_followup` can have the description "Systolic blood pressure at the three-month visit, mmHg." A clear name helps identify the variable in analysis dialogs, while the description preserves its definition.

2.5.1 Illustrative import checklist

Check	What to verify
Dimensions	Expected number of records and variables
Unit of observation	One patient, visit, admission, or other defined unit per record
Names	Unique, meaningful column names
Numeric variables	Values are numeric; units are documented separately
Categories	Codes and labels match the data dictionary
Missing values	Blank or coded missing entries are distinguished from valid zero values
Identifiers	Record linkage and duplicates are understood
Group structure	Independent groups or paired measurements match the intended analysis

2.6 Variable properties and levels

Check the measure type after importing. Imported values can be stored as integer, decimal, or text, while their measure type may be nominal, ordinal, continuous, or ID. jamovi may infer properties on opening a file, so the researcher should verify them against the dictionary. [4]

An outcome such as blood pressure should retain its measurement unit and continuous meaning. A grouping variable should display its category labels. A participant identifier should link records rather than act as a predictor solely because it contains numbers. For an ordinal variable, inspect the order of the levels as well as their labels.

Reference categories require particular care. If an analysis reports group B relative to A, the sign or ratio direction differs from A relative to B. Write the intended comparison before reading the output. Later chapters show how this direction affects mean differences, correlation, and odds ratios.

2.7 Saving data and analyses

A plain data file and a saved analysis file serve different purposes. A CSV can provide an accessible rectangular data source, but it does not preserve the complete set of jamovi analyses and options. The `.omv` format saves the jamovi project information needed to reopen the analysis. [7]

Keep three related items: the unchanged source file, a documented analysis-ready file, and the saved jamovi analysis. Give each an understandable name and version date. For example, an author may distinguish `clinic_source`, `clinic_clean_v01`, and `clinic_analysis_v01`; the exact naming convention is less important than its consistent use.

When sharing a result for review, include the analysis file, data dictionary, and a short record of exclusions and transformations. A screenshot can help explain a table, but another researcher should be able to examine the underlying variables and options as well.

2.8 Reviewing an output before interpretation

Read the analysis title and verify that it matches the intended method. Check the variables and sample size. Identify the estimate and its direction. Locate the confidence interval, test statistic, degrees of freedom where applicable, and p-value. Check any note about missing values, assumptions, or estimation failure.

Suppose a table contains 80 observations although the imported file has 100 records. Possible explanations include missing values, an active filter, or an analysis limited to one subgroup. Determine which explanation applies before interpreting the estimate. The difference between imported records and analysed observations must be understandable.

Do not copy every software table into a manuscript. Retain the output for verification, then choose the quantities needed to answer the research question. A reader-facing table has a defined purpose and usually requires clearer labels than those used in the analysis window.

2.9 R as a later reproducibility layer

jamovi offers syntax mode and Rj for working with R. These provide routes to code-based analysis, with package and environment requirements that should be checked when they are used. The availability of R integration does not mean that every method proposed for the series is supplied by the default interface. [9]

For Volume I, begin with transparent data definitions, saved analyses, and recorded options. The project's verification script separately reproduces the small numerical illustrations using base R; it is not a replacement for the author's future jamovi demonstrations. Later volumes develop a fuller relationship between graphical analysis and code.

2.10 Hands-on

jamovi workspace

2.11 Review questions

1. Why should a numeric participant identifier not automatically be treated as continuous?
2. What can an `.omv` file preserve that a plain CSV does not preserve?
3. An output uses fewer observations than the imported file. What should be checked before interpreting it?

2.12 Answers and reasoning

1. Its numbers are labels for linking observations; arithmetic on them does not answer the health question.
2. It preserves the jamovi analysis context, including analyses and their settings, alongside the data.
3. Check missingness, active filters, subgroup selections, and the analysis-specific requirements. Report the resulting analytical denominator.

Chapter 3

Data Entry and Validation

3.1 Learning objectives

After this chapter, the reader should be able to define what one row represents, build a usable data dictionary, distinguish data type from measure type, specify entry checks, and verify that an imported file retains its intended values and structure.

3.2 Start with the measurement plan

Data entry is the conversion of recorded observations into a structured analytical file. Validation asks whether that conversion follows the study's definitions. A file can open successfully and still contain the wrong unit, a truncated identifier, or a category whose code has changed meaning. Successful import is therefore the beginning of validation.

Before entering data, identify the source, population, observation unit, measurement occasion, and expected number of records. Specify who records each item and how unclear source entries will be resolved. The questionnaire, case record form, and data dictionary should use compatible definitions. An analysis cannot distinguish a missing measurement from an unrecorded negative result if the collection form did not preserve that distinction.

The four preparation chapters have separate responsibilities. This chapter covers capture and validation. Chapter 4 addresses verified corrections and missingness. Chapter 5 develops derived variables. Chapter 6 assembles the variables, eligible observations, and settings needed for a particular analysis.

3.3 Define the row and its identifier

A row may represent a participant, visit, admission, specimen, household, or event. State its meaning before choosing an identifier. A participant-level file ordinarily needs a unique participant ID. A visit-level file may need the combination of participant ID and visit occasion to identify a row uniquely.

An identifier links records; its numerical appearance does not make it a quantitative measurement. The ID `0017` may be different from `17` in a source system. Preserve the intended format during entry and import. Do not use the current row number as a substitute for a stable identifier, because sorting or inserting records can change that number.

For paired measurements, the same row must contain measurements from the same participant, or a verified key must link the separate files. A convenient row order does not establish matching.

For clustered data, retain the cluster identifier even when the current volume's simple methods cannot accommodate the full dependence structure.

3.4 Build the data dictionary

The dictionary should allow another researcher to interpret a column without asking the person who entered it. Include the name, definition, source, timing, unit, data and measure types, permitted values, missing codes, and any validation rule. A derived variable also needs its formula and the version of its source variables.

The following entries are invented specifications for a teaching study:

Variable	Meaning and timing	Representation	Validation rule
`study_id`	Participant code assigned at enrolment	Text ID	Present and unique in the participant file
`group`	Defined study group	Nominal; A or B	One permitted label per eligible participant
`sbp_baseline`	Baseline systolic blood pressure	Decimal; continuous; mmHg	Numeric when observed; source checked when flagged
`sbp_followup`	Value at the specified follow-up visit	Decimal; continuous; mmHg	Timing meets the study's visit rule
`event`	Defined event during the stated period	Nominal; 1 present, 0 absent	Unknown status is missing, with a reason recorded separately
`severity`	Recorded ordered category	Ordinal; mild, moderate, severe	Order matches the collection form

The blood pressure entries intentionally contain no invented clinical acceptance limit. A project's limits should come from its measurement protocol and instrument, and should distinguish an error from a value that needs review.

Use short, stable variable names and informative descriptions. Keep units out of the cells containing numeric values: store `128`, with mmHg in the dictionary, rather than mixing `128 mmHg` and `128` in one column. Store a categorical response in its own column instead of using cell colour to encode its meaning. These principles are consistent with recommendations for organising spreadsheet data. [10]

3.5 Separate data type from measure type

The **data type** describes how values are stored. jamovi distinguishes integer, decimal, and text data. The **measure type** describes how a variable should be treated: continuous, ordinal, nominal, or ID. The automatic choices made during import need review against the dictionary. [4]

A study group coded 1 and 2 is nominal, because subtracting the codes has no scientific meaning. A count stored as an integer may be used as a quantitative variable, but that storage choice does not make every statistical model appropriate. An ordinal severity score retains order without establishing equal distances between categories.

Check category labels as well as codes. If 1 means present in one source and absent in another, the same stored value has different meanings. Establish a common mapping from the dictionaries before combining files. Record the reference group and ordinal order explicitly; they affect the presentation and interpretation of later results.

3.6 Choose a consistent table structure

Keep one header row, one variable per column, and one value per cell in a rectangular entry table. Put explanatory notes, totals, and presentation tables elsewhere. Avoid merged cells, repeated headers within the data, and blank rows used to separate groups. They describe a visual layout rather than an analytical structure. [10]

Wide format places different measurement occasions in separate columns for the same participant. This is a useful starting structure for the simple paired procedures in Volume I. **Long format** places occasions in separate rows and preserves an ID and time variable. Neither arrangement removes dependence between measurements from one participant.

Intended question	Starting structure	Required check
Independent group comparison	Participant ID, outcome, group	Each participant contributes once
Baseline and follow-up comparison	Participant ID and both measurement columns	The columns form valid within-person pairs
Repeated visit description	Participant ID, visit, outcome	ID plus visit identifies the row; counts distinguish people from visits
Event proportion over a defined period	Participant ID and event status	Eligibility, observation period, and unknown status remain explicit

Keep the source structure when it contains information that a later analysis needs. Reducing repeated visits to one outcome is an analytical decision with a selection rule, rather than a routine formatting operation.

3.7 Use checks at entry and after entry

Validation rules should be specified before viewing the statistical results. Their role is to detect departures from the collection definitions, rather than force the dataset to fit a preferred distribution.

Check	Example of what it detects	Required response
Required field	Missing participant identifier	Resolve from the source before linkage
Permitted category	Group recorded as an undefined C	Query the source or dictionary
Data format	Text in a numeric measurement	Check for units, special codes, or import errors
Range or plausibility	Value outside a protocol-defined range	Distinguish an error from a valid unusual value
Cross-field consistency	Follow-up recorded before enrolment	Verify dates and the occasion definition
Uniqueness	Repeated participant key	Determine whether it is duplication or a different observation unit

A **hard rule** concerns a contradiction with the definition, such as a negative count of events. A **review rule** flags a value whose validity requires evidence, such as an unusually high measurement. A flag is not itself an instruction to delete the observation. Keep the rule, source query, and eventual decision traceable.

Some entry systems allow limits and permitted categories to be enforced immediately. A spreadsheet or imported file still needs a separate validation pass. Independent double entry or a targeted source check may be appropriate when the error consequences justify the effort. Record the verification coverage; checking a sample does not establish that every value was checked.

3.8 Distinguish missing, zero, and not applicable

Zero is a value. Missing means that the intended value is unavailable. Not applicable means that the item does not apply under a defined rule. An event count of zero among observed participants is different from an unknown event count, and both differ from a measurement that was never required.

Keep a numeric measurement column suitable for calculation and record missing reasons in a separate field when those reasons matter. A reason might distinguish an absent visit, a failed measurement, and a participant outside the eligible measurement group. Do not use a numeric zero for all three situations.

jamovi represents missing values as blank cells. If a source uses a special code, define that code as missing in the variable settings or create a documented cleaned variable before analysis. Otherwise the code may be treated as an ordinary value. [4] A special code is missing because the dictionary says so, rather than because it is inconvenient for the analysis.

3.9 Verify the import

Preserve the source file and import a named working copy. Note the source version, export format, delimiter, decimal convention, and date convention. Inspect whether quoted text, accented labels, leading zeros, and blank cells have survived the transfer. If a date such as `05/06/2026` is ambiguous, resolve its convention from the source rather than from whichever interpretation produces a plausible chronology.

Compare expected and imported records, variables, and category counts. Inspect a few source values from the beginning, middle, and end of the file, including values that challenge the format: a decimal, a special missing code, an identifier with leading zeros, and a label containing punctuation. Reconcile discrepancies before calculating results.

An imported numeric column may be text if some cells include units or unexpected characters. Converting the whole column without investigating those cells can create new missing values. Document the number of conversions that succeed and fail, and verify that no genuine observations disappeared unnoticed.

3.10 An invented validation audit

Suppose an entry file is expected to contain 120 participants and 12 specified variables. It imports with 120 rows and 12 columns. A further audit identifies two rows sharing an ID, four uses of the undeclared group label `b`, and three follow-up fields containing the dictionary's missing code 999.

The matching row and column counts do not establish validity. Query the repeated ID before deciding whether it represents a duplicate or an incorrect identifier. Confirm whether `b` means group B before harmonising it. Define 999 as missing according to the dictionary and retain the reason if available. The issues may overlap within participants, so adding 2, 4, and 3 does not necessarily count the number of affected people.

Write the audit finding before correcting the working file. The next chapter explains how to resolve and log the findings. Validation discovers an issue; cleaning implements a supported decision.

3.11 Release a validated entry file

Before handover, confirm the observation unit and keys, reconcile record counts, verify variable properties, inspect categories and missing codes, and list unresolved queries. Save the dictionary with the entry file. A reviewer should know whether the file is ready for cleaning or whether source questions remain.

For reporting, describe the collection and validation process at the level needed to assess the data. A statement such as “Data were checked” is insufficient when the relevant questions concern matching, eligibility, or conflicting measurements. The later reporting chapter explains how these preparation decisions enter the methods section.

3.12 Hands-on

jamovi workspace

3.13 Review questions

1. Why can 120 imported rows fail to represent 120 distinct participants?
2. A group is coded 1 and 2. What determines its measure type?
3. Why should an unusual value trigger review rather than automatic deletion?
4. Which checks are needed when an ID such as 0017 is imported?

3.14 Answers and reasoning

1. A row may be duplicated, may represent a repeat visit, or may contain an incorrect ID. Count participants using a verified key and the defined observation unit.
2. The scientific meaning determines the measure type. Group membership is nominal even when its stored codes are numbers.
3. An unusual observation can be valid. Resolve it against the source and protocol, and retain a record of the decision.
4. Verify the intended text format, leading zeros, uniqueness, and linkage to the source; do not assume the numeric-looking value has survived unchanged.

Chapter 4

Data Cleaning and Missing Values

4.1 Learning objectives

After this chapter, the reader should be able to resolve a data query using supporting evidence, distinguish duplicates from repeat observations, document corrections, describe missingness with appropriate denominators, and explain the limits of a complete-case analysis.

4.2 Cleaning is a documented decision process

Cleaning changes an analytical representation when an issue has been identified and a supported resolution is available. It includes correcting a transcription error, harmonising equivalent labels, recognising special missing codes, and resolving duplicated records. Each action needs a reason that another researcher can inspect.

Keep the original export unchanged and work on a named copy. Record the source version and the dictionary used to interpret it. Save the rules and change log with the cleaned file. Preserving a raw file separately from analytical work is part of good spreadsheet organisation. [10]

The aim is to represent the recorded observations correctly. A cleaner distribution, a larger analytical sample, or a smaller p-value is not evidence that a correction is valid. When a finding cannot be resolved, retain its status and explain how it will affect the analysis.

4.3 Classify the finding before changing the value

Separate confirmed errors, equivalent representations, valid unusual observations, and unresolved queries. They require different responses.

Finding	Evidence needed	Suitable action
Transcription error	Source confirms a different value	Correct the working value and log the change
Equivalent category labels	Dictionary confirms a common meaning	Apply a consistent label mapping
Special missing code	Dictionary defines the code	Represent it as missing for calculation
Valid unusual observation	Source and measurement definition support it	Retain it; assess influence if relevant
Unresolved conflicting values	Available sources do not settle the discrepancy	Record the query and apply a stated analytical rule

Do not guess that a large value is a misplaced decimal point. Query the source. If it confirms the recorded value, the statistical analysis must accommodate or transparently assess it. If it confirms an error, correct that error because of the source evidence, even if the result is almost unchanged.

4.4 Resolve duplicates using the correct key

In a participant-level file, an ID repeated twice is a query. In a visit-level file, the same ID can appear legitimately at several occasions. The combination of ID and visit may be the correct row key. Check the observation unit defined in Chapter 3 before removing anything.

An exact duplicate has the same relevant values and key as another record. A conflicting duplicate shares a key but contains different values. The latter requires source review; retaining the first or last row by default can select an arbitrary measurement.

Suppose an invented file contains 204 rows expected to represent one row per participant. Source review establishes that four rows are repeated exports of existing records, leaving 200 distinct participant records. This arithmetic is valid only after confirming what those four rows represent. A file with 204 legitimate visits from 200 participants would require a different interpretation.

If several measurements are eligible, use a defined selection rule, such as a protocol's scheduled occasion and allowed time window. Record tie handling. Averaging repeated measurements or choosing one is a derivation with consequences for the target quantity; it is not merely duplicate removal.

4.5 Harmonise categories without changing their meaning

Inspect frequency tables before and after recoding. Differences in case, spacing, or spelling may produce separate labels for the same category. Confirm equivalence from the dictionary or source, then map them to a defined common label.

For example, `B`, `b`, and `Group B` may represent the same study group. `Unknown` and `No` do not represent the same event status. A text value such as `NA` is also ambiguous until the source explains whether it means missing, not applicable, or a legitimate category.

Preserve the original label when the mapping needs an audit. Compare original and cleaned variables in a cross-tabulation. Verify that every intended category has a destination, and inspect values left unmapped. A new label alone cannot demonstrate that the recode worked.

4.6 Check units, dates, and internal consistency

A measurement can be numerically plausible while using the wrong unit. Review the source unit and dictionary before conversion. When a column combines values measured in centimetres and metres, an additional unit field or reliable source record is needed to determine which conversion applies. Guessing from the magnitude can misclassify valid observations.

Check chronology against the study definitions. A recorded follow-up before baseline can indicate an entry error, an incorrect visit label, or a source-system convention. Confirm the explanation before calculating elapsed time. Preserve the original date representation and specify the parsing rule used in any external preparation step.

Cross-field checks should reflect the collection process. An event date outside the defined observation period is a query about the outcome definition. A total that differs from the sum of recorded components is a query about calculation or measurement. Record the resolved values and the evidence, rather than silently forcing consistency.

4.7 Distinguish missingness from exclusion

An ineligible participant is outside a stated analysis population. A participant with a missing outcome may be eligible but lack the value needed for a particular calculation. A verified duplicate is an extra representation of an existing observation. These are different reasons for changing a record count.

Retain reason fields so the counts can be reported separately. If several exclusion criteria overlap, use a stated order for a sequential flow and also retain the underlying flags. Summing overlapping flags does not necessarily equal the number excluded.

Missingness should be described among the observations for which the measurement was intended. Report the eligible number, observed number, and missing number. If not-applicable observations are excluded from that denominator, explain the rule.

4.8 An invented sample-flow example

The following counts illustrate documentation; they do not describe an included patient dataset. Begin with the 200 distinct participant records from the duplicate example. Eight fail the specified eligibility criteria, leaving 192 eligible participants. Among these, 12 lack the follow-up outcome, leaving 180 available values for that outcome.

Stage	Number	Meaning
Imported rows	204	Includes four verified duplicate rows
Distinct participants	200	Duplicate resolution completed
Eligible participants	192	Eight distinct participants are ineligible
Observed follow-up outcomes	180	Twelve eligible participants lack this outcome

The missing-outcome percentage among eligible participants is $12/192 \times 100 = 6.25\%$. The 180 observed outcomes represent 93.75% of eligible participants. The missing proportion is not 12/204, because imported rows are not the relevant eligible participant denominator.

Suppose the eligible groups each contain 96 participants. Four outcomes are missing in A and eight in B, leaving 92 and 88 observed values. The group-specific missing proportions are 4.17% and 8.33%. A similar overall percentage could hide a different pattern between groups.

These counts can describe the missingness pattern. They cannot establish that the missing outcomes would have the same distribution as observed outcomes. Interpretation requires information about follow-up and reasons for absence.

4.9 Understand assumptions about missingness

Missing completely at random (MCAR) means that the missingness process is unrelated to observed and unobserved values. **Missing at random (MAR)** means that, conditional on relevant observed information, missingness does not depend on the missing values. **Missing not at random (MNAR)** means that dependence on unobserved values remains. These are properties and assumptions of the data process, rather than conclusions obtained from the percentage of blank cells. [11]

A missed visit caused by an administrative accident may have a different relationship with health status from a missed visit caused by worsening illness. Even when a reason is recorded, its implications require judgment. Observed associations with missingness can challenge a simple assumption, but absence of such associations does not establish that unseen values are irrelevant.

Record known reasons, examine missingness by group or occasion, and identify which assumptions the intended analysis needs. Formal missing-data modeling and sensitivity analysis belong beyond the introductory workflow in this volume.

4.10 Know what complete-case analysis does

A complete-case analysis includes observations that have the values needed for its specified variables. A follow-up group comparison may use 180 outcomes in the invented example, while a paired analysis may use fewer if some baseline values are also missing. Missingness in an unrelated variable need not exclude a participant from the first calculation.

R's `complete.cases` identifies observations without missing values in the supplied variables. The variables supplied therefore determine the case definition. [12] jamovi analyses also offer different missing-data handling choices in some procedures; inspect the relevant analysis settings and the resulting sample size. [13]

Complete-case analysis changes the analysed set and may change the population represented by the result. Its validity depends on the missingness process, design, and target quantity. A large retained sample is not sufficient proof that bias is negligible.

State which variables defined completeness, how many observations were omitted, and what is known about their missingness. If the same analysis is compared across different models, explain whether the sample changed as well as the model.

4.11 Why simple replacement is not a neutral repair

Replacing an unknown value with zero turns absence of information into an observed measurement. Replacing every missing outcome with the sample mean creates repeated invented values and suppresses the variation that those outcomes might have contained. Neither action restores the lost information.

Carrying a previous measurement forward also imposes a substantive assumption about change. It should not be described as an ordinary cleaning step. An imputation procedure needs a rationale, an appropriate uncertainty calculation, and assumptions that match the study. [11]

For this introductory workflow, resolve coding errors, preserve actual missingness, report the analytical sample, and acknowledge the limitation. Seek an appropriate missing-data strategy when the intended research conclusion depends materially on unavailable values.

4.12 Handle unusual observations transparently

Use summaries and graphs to locate values for review, but use evidence to decide whether they are errors. A boxplot flag or a large standardised distance is a description of an observation relative to the sample, not proof that the observation is invalid.

A confirmed unusual value can have substantial influence on a mean or fitted line. An additional analysis examining that influence may be informative when justified and labelled. Report the main analysis rule and the sensitivity result; do not replace the primary result silently because an alternative is more convenient.

Deleting valid extreme values changes the analysed distribution and can change the target population. An exclusion rule should follow the design or a defensible measurement rule and should be stated before selecting results.

4.13 Keep a change log and a release record

A useful log includes the file version, affected key or variable, issue, supporting source, action, reason, date, and reviewer. Keep identifying details limited to what the authorised review process requires.

Log field	Example entry for a teaching file
Issue	Three records use lowercase group label `b`
Verification	Dictionary and source confirm group B
Action	Map `b` to B in the cleaned group column
Check	Original-by-cleaned table has no unintended mapping
Record	Rule, affected count, date, and reviewer retained

Before release, reconcile row and participant counts, repeat range and category checks, compare missingness with the raw file, and list unresolved issues. A corrected missing code may increase the recognised missing count even though no information was lost during cleaning. Explain that change.

The cleaned file is ready for derivation when its values and structure have been reviewed. It is ready for a particular analysis only after the population, required variables, matching, and exclusion settings have also been checked in Chapter 6.

4.14 Hands-on

jamovi workspace

4.15 Review questions

1. Why is a repeated participant ID not always a duplicate row?
2. In the invented flow, which denominator gives the missing follow-up percentage?
3. Does 6.25% missingness establish MCAR or MAR?
4. Why may a paired analysis contain fewer observations than a follow-up-only comparison?

4.16 Answers and reasoning

1. Rows may represent different visits or events. The observation unit and full row key determine whether repetition is legitimate.
2. The 192 eligible participants for whom follow-up was intended; 12/192 is 6.25%.
3. No. Those assumptions concern how missingness relates to observed and unobserved values, which a percentage cannot establish.
4. The paired calculation needs both defined measurements. An observed follow-up without baseline does not form a complete pair.

Chapter 5

Transformations and Derived Variables

5.1 Learning objectives

After this chapter, the reader should be able to specify a derived variable, preserve its source measurements, distinguish row calculations from column summaries, check recoding boundaries and missingness, and explain how a transformation changes the scale of an analysis.

5.2 Define the purpose before the formula

A derived variable expresses a quantity needed for the research question. A change score, a unit conversion, an ordered category, and a logarithm are different operations with different meanings. Write the intended definition in words before implementing its formula.

Keep the source columns. Give the derived column a distinct name and a description containing its unit, direction, formula, permissible inputs, and missing-value rule. Record why the derivation is needed and whether it was specified before examining the outcomes.

Cleaning resolves a supported representation problem. Transformation creates a new representation for an analytical purpose. Correcting a verified transcription error is cleaning; calculating follow-up minus baseline is derivation. A mathematical transformation does not repair invalid source values or make an unsuitable study design valid.

5.3 Computed and transformed variables in jamovi

jamovi's computed variables create columns from formulas involving existing variables. Transformed variables allow a reusable rule to be applied to source columns. A transformation refers to its selected source with `$source`, and recode conditions are evaluated in order, using the first condition that is true. [14, 15]

The chapter uses small formula illustrations. They are not results from an included dataset or a tested saved jamovi analysis. The author will add version-specific demonstrations and screenshots in the reserved Hands-on section.

Choose the tool according to the derivation. A difference between two specified columns is naturally a computed variable. Reversing several items on the same response scale can use one reusable transformation. In either case, the formula needs independent checks before it supports a reported result.

5.4 Distinguish calculations within a row from column summaries

A row calculation uses values belonging to one observation. A column summary uses a set of observations. Confusing these levels can produce a syntactically valid but scientifically wrong variable.

For example, the mean of three specified questionnaire items within a participant differs from the mean response to one item across participants. Likewise, an individual's change from baseline differs from the difference between two column means when the columns contain different observed participants.

The documented jamovi function names distinguish several row and variable operations: `MEAN(...)` and `SUM(...)` are row functions, whereas `VMEAN(var)` and `VSUM(var)` summarise a column. `Z(var)` standardises a variable, and `LN(var)` takes natural logarithms. [16] Check the function's input and missing-value behaviour against the intended rule; a function name alone does not define a valid score.

5.5 Unit conversion and ratios

Convert only when the original unit is known. A height of 160 cm becomes 1.60 m by dividing by 100. Retain the source height and label the converted column in metres. Do not combine units in one column and infer the unit solely from magnitude.

Body mass index uses weight in kilograms divided by squared height in metres. [17] An invented illustration uses a weight of 80 kg and a height of 1.60 m:

$$\text{BMI} = \frac{\text{weight in kg}}{(\text{height in m})^2} = \frac{80}{1.60^2} = 31.25 \text{ kg/m}^2.$$

This is a derivation example, without a clinical classification. Inserting 160 instead of 1.60 gives 0.003125, showing why unit verification must precede calculation. A zero, missing, or unresolved height cannot produce a valid BMI; investigate the source rather than treating an infinite or absent result as a new measured value.

A ratio also requires a meaningful denominator. A rate per unit time requires observed time in the stated unit. When the denominator is zero or unknown, define how the result is represented and what the source problem means. Do not silently convert an undefined ratio to zero.

5.6 Change scores and direction

For paired baseline and follow-up measurements, define

$$d = \text{follow-up} - \text{baseline}.$$

With this definition, negative values indicate a lower follow-up measurement. The formula illustration `sbp_followup - sbp_baseline` uses columns belonging to the same participant. Verify matching before calculating it.

For invented baseline and follow-up values of 146 and 138 mmHg, the change is -8 mmHg. Reversing the subtraction gives +8 mmHg. The statistical evidence can be equivalent under a consistent sign change, but the reporting language must follow the stated direction.

An individual's percentage change, when baseline is a valid non-zero denominator, is

$$100 \times \frac{\text{follow-up} - \text{baseline}}{\text{baseline}}.$$

For those values it is -5.48%. Percentage change and absolute change answer different questions. A smaller baseline can produce a larger percentage change for the same absolute difference. When baseline is zero, the ratio is undefined. When baseline is missing, no observed within-person percentage change exists.

The average of individual percentage changes is also generally different from the percentage change in the two group means. Define which quantity is being reported. Preserve enough information to reproduce that calculation.

5.7 Recode categories with explicit boundaries

Recoding needs a complete mapping, not just a new name. Specify the destination for each permitted source category and the treatment of missing or unexpected values. Verify the mapping with an original-by-derived table.

For an invented waiting-time categorisation, define three groups without implying a clinical threshold:

Source value, minutes	Derived category	Boundary check
$0 \leq x < 10$	Shorter	0 belongs here; 10 does not
$10 \leq x < 20$	Intermediate	10 belongs here; 20 does not
$x \geq 20$	Longer	20 belongs here
Missing or invalid input	Missing or flagged for review	Do not label it as a valid shorter wait

Check values just below, exactly at, and just above each boundary. Mutually exclusive conditions avoid ambiguity. If a procedure uses the first true rule, a broad condition placed first can make later conditions unreachable. [15]

Categorisation removes distinctions within each category. Values of 10 and 19.9 become identical under the invented middle category, while 19.9 and 20 become different despite their small separation. Dichotomising a continuous variable can discard information and reduce statistical power. [18]

Use a category when its definition serves the question, with a stated source or purpose. Retain the original continuous variable. Do not search repeatedly for the threshold that produces the preferred result.

5.8 Reverse scoring and composite scores

For an item coded from a minimum a to a maximum b, reversal is

$$x_{\text{reversed}} = a + b - x.$$

On an invented 1-to-5 response scale, this becomes $6 - x$: 1 maps to 5, 2 to 4, and 3 remains 3. A reusable jamovi transformation can express this as `6 - $source`. [15] Recognise declared missing codes before reversal; a missing code of 99 should not become a score of -93.

Reversal is justified by the scoring specification, not by whether it improves an internal consistency statistic. For an established instrument, follow its permitted item coding, scoring, and

missing-item rules. Correct arithmetic does not establish that an invented item set measures a valid construct.

Suppose three invented items, after correct reversal where required, have scores 4, 5, and 3. Their sum is 12 and their mean is 4. If one item is missing, a sum of only the two available items is not comparable with a complete three-item sum. An available-item mean also makes assumptions about what the missing item represents.

State the minimum number of required items and the procedure when that minimum is not met. A prorated or available-item score is a deliberate rule that needs justification; it should not arise accidentally from a function's default handling of blanks. Verify complete, partly missing, and entirely missing examples separately.

5.9 Logarithms change the scale of the question

The natural logarithm is defined for positive values. For a positive variable x, the transformed value is $\ln(x)$; the original scale is recovered with $\exp(\ln(x)) = x$. The documented jamovi function `LN(var)` performs the natural-log calculation. [16]

On the log scale, equal ratios give equal differences. Increasing from 10 to 20 and from 20 to 40 gives the same log difference, $\ln(2) = 0.6931$, despite different absolute increases. This can be useful when multiplicative relationships match the scientific question.

Transforming individual values also changes the meaning of an average. For the invented positive values 10 and 40, the arithmetic mean is 25. The mean of their natural logs, back-transformed, is 20, the geometric mean. It is not a new way to calculate the arithmetic mean.

Zero and negative values need substantive attention before a log transformation. Adding a constant merely to make the calculation possible changes the resulting scale and interpretation. Adding 1 in one measurement unit need not have the same effect after converting to another unit. Explain the chosen definition rather than treating it as a universal repair.

A transformation may improve the suitability of a model for its intended relationship, but it should not be selected only to pass a normality test. Inspect the transformed distribution and, when modeling, the relevant residual behaviour. Later chapters connect those checks with the method's assumptions.

5.10 Standardisation and its reference set

A sample z-score is

$$z_i = \frac{x_i - \bar{x}}{s},$$

where the mean and SD refer to a stated set of observations. For an invented value of 140 mmHg, reference mean 130, and reference SD 10, the z-score is 1. The score describes relative position in that reference distribution; it does not define a clinical diagnosis.

Record whether the reference comes from all eligible observations, a group, or an external population. If the reference mean or SD changes, the same measurement receives a different score. A column with zero SD cannot be standardised by this formula.

Active filters can matter for calculations involving column summaries. jamovi's filter documentation describes how sequential filters affect the observations used in subsequent calculations. [19] Verify the actual reference set when saving standardised variables. A value derived under one filter state should not be described as based on a different sample.

5.11 Verify and document the derivation

Check simple values by hand, both boundaries of recodes, missing inputs, invalid inputs, and the range and distribution of the new variable. Compare selected derived values with an independent calculation. Reconcile the number of missing results with the rule rather than assuming every new blank is expected.

For a change score, confirm direction and matching. For a unit conversion, confirm the factor and output unit. For a category, inspect the mapping. For a score, check item reversal and missing-item eligibility. For a log, count non-positive inputs before interpreting the transformed values.

Keep full useful precision in the analytical column and round for presentation. Rounding before deriving a boundary category or a difference can change the classification or result. Record the transformation definition with the dictionary so a reviewer can rebuild it from the source variables.

5.12 Hands-on

jamovi workspace

5.13 Review questions

1. Why can follow-up minus baseline differ from the difference between the two available column means?
2. Under the invented waiting categories, where do 10 and 20 minutes belong?
3. Why should 99 be recognised as missing before reversing a 1-to-5 item?
4. Is the back-transformed average of log values the arithmetic mean?

5.14 Answers and reasoning

1. Individual differences require complete matched measurements. Available column means can use different participants when missingness differs.
2. Ten belongs to Intermediate and 20 to Longer; the boundaries were defined explicitly.
3. Applying 6 - 99 would create an invalid score of -93 instead of preserving the declared missingness.
4. No. Averaging logs and exponentiating gives a geometric mean for positive values; 10 and 40 illustrate 20 versus the arithmetic mean of 25.

Chapter 6

Preparing Data for a Specific Analysis

6.1 Learning objectives

After this chapter, the reader should be able to write an analysis specification, select the variables and observations needed for a basic method, preserve pairing and denominators, check active filters and missing-data rules, and document the final analytical sample.

6.2 A cleaned file does not define every analysis

A reviewed dataset can support several questions, each using different variables and observations. Describing a follow-up measurement, comparing paired measurements, and fitting a relationship with age do not necessarily use the same sample. Analysis preparation makes those differences explicit before interpreting results.

The process starts with the research question and target quantity. State who is eligible, what one observation represents, which outcome is being analysed, which comparison or predictor is relevant, and at what time the outcome is measured. Then select a method that matches that specification.

Do not begin by trying every procedure that accepts the columns. The software's acceptance of a variable does not establish that the variable has the right role, that observations are independent, or that the result answers the intended question.

6.3 Write a short analysis specification

The specification connects the design to the analytical file. It can be brief, but it must contain enough information to reproduce the included observations, variable definitions, method, and contrast direction.

The following specification is invented for an unadjusted follow-up comparison:

Component	Stated decision
Question	How does mean follow-up systolic blood pressure differ between eligible groups A and B?
Outcome	The protocol-defined follow-up measurement, in mmHg
Observation unit	One eligible participant, contributing once
Comparison	A minus B

Component	Stated decision
Analytical sample	Eligible participants with known group and observed follow-up outcome
Method	Welch independent-samples comparison, if the design and data support its use
Descriptive output	Group-specific n, mean, SD, missing count, and distribution plots
Main result	Mean difference with 95% CI, followed by the test result
Limitation	An unadjusted comparison does not account for baseline or other group differences

This specification does not assert that the assumptions have already passed. It identifies what must be checked and what the result will mean. Where the study requires a baseline-adjusted or other multivariable analysis, the simple comparison is insufficient as the final model; those methods are developed in Volume II.

Distinguish a prespecified primary analysis from an exploratory analysis. If the plan changes because of a data issue, record what changed and why. Do not describe a choice made after inspecting several results as if it had been fixed beforehand.

6.4 Match the variables and structure to the method

Check the scientific definition, measure type, categories, and required matching for each selected variable. The following map supports the core methods in this volume:

Method	Required variables	Observation check
Descriptive summary	Variable of interest; optional group	Correct denominator for each reported summary
Independent two-group comparison	Quantitative outcome and two-level group	Participants contribute once to one group
Paired comparison	Two defined measurements or a within-pair difference	Valid pairs, with independence between pairs under the simple model
One-way group comparison	Quantitative outcome and defined group levels	Independent observations under the basic model
Categorical association	Two categorical variables	Count participants in the correct cells; check design and denominator
Correlation	Two quantitative or appropriately ordered variables	Values belong to the same observation; confirm paired completeness
Simple linear regression	Quantitative outcome and one defined predictor	Correct outcome role and defensible mean relationship
Introductory logistic regression	Defined binary outcome and one predictor	Event coding and reference group are explicit

This table is a preparation map, not a complete test-selection rule. The relevant chapters explain assumptions, alternative targets, and reporting. Appendix B provides the method-selection guide.

6.5 Preserve the unit of observation

A file with one row per visit can contain more rows than participants. An independent t-test applied to all those rows would ordinarily fail to account for the relationship between visits from

the same person. A matching ID does not make the ordinary independent procedure handle that dependence automatically.

Specify whether the question concerns participants, visits, or another unit. If one measurement per participant is required, select it using a justified occasion rule and document how conflicts are resolved. If repeated observations are part of the question, use a method that models their structure rather than deleting the dependence from the description.

Similarly, participants sampled within households or clinics may share cluster-level influences. Preserve the cluster variables and identify when the simple independent methods in this volume are insufficient. A larger row count cannot compensate for an inappropriate independence assumption.

6.6 Define eligibility before activating filters

An eligibility rule determines which observations belong to the target analysis population. Write it in words and implement it using defined variables. Separate eligibility from the availability of measurements needed for a particular calculation.

For example, first identify eligible participants, then determine which have the follow-up outcome. Record both counts. If an unknown eligibility field is encountered, resolve it or state the handling rule; treating an unknown value as an automatic yes or no can change the target population.

jamovi row filters exclude observations from analyses without requiring deletion of the source records. The filter state can therefore change existing results. Multiple filters and calculations involving column summaries need attention to their order and reference set. [19]

Check every active filter when the displayed sample size is unexpected. Record its expression, purpose, and included and excluded counts. An eligibility filter for one question may be inappropriate for another question in the same file. Save separate analysis files or clearly documented filter states when necessary to avoid confusing their samples.

Do not hide an analytical exclusion inside an unexplained derived column. If a value has been made missing to implement a subset, that is part of the inclusion rule and must be recorded as such.

6.7 Account for missing values for the selected variables

The required-variable list defines what completeness means. A descriptive summary of age does not necessarily require an observed follow-up outcome. A paired blood pressure comparison requires both defined measurements. A correlation requires an observed value of each variable for the same participant.

Inspect the sample count reported by each analysis and reconcile it with the eligibility and missing-value rules. jamovi's one-way documentation, for example, distinguishes per-analysis and listwise exclusion options. [13] Different procedures and options should not be assumed to use identical cases.

When comparing models or tables, state whether differences arise from the method, the selected variables, or the analysed participants. Holding the sample constant can sometimes help isolate a model comparison, but it also excludes observations and needs a stated purpose. It is not an automatic requirement for every descriptive result.

6.8 An invented complete-pair audit

Consider a separate teaching example with 120 eligible participants. Six have no baseline measurement, eight have no follow-up measurement, and two lack both. There are 114 available baseline values and 112 available follow-up values, but neither number gives the complete-pair count.

The number lacking at least one measurement is $6 + 8 - 2 = 12$, because the two who lack both were counted twice in the sum. The paired analysis therefore has $120 - 12 = 108$ complete pairs.

Availability	Number	Contribution to a paired calculation
Both measurements observed	108	One defined difference per participant
Baseline missing only	4	No complete difference
Follow-up missing only	6	No complete difference
Both missing	2	No complete difference

For these 108 pairs, the mean of individual follow-up-minus-baseline differences equals the follow-up mean minus the baseline mean calculated on those same pairs. It need not equal the difference between the means calculated from all 112 follow-up and all 114 baseline values.

This identity depends on using the same matched participants. It explains why matching and missingness checks precede a paired test. The numerical illustration in the paired-comparison chapter is a different teaching case with 30 complete pairs.

6.9 Confirm categorical denominators and event coding

For an event proportion, state the observation period, event definition, eligible population, and number with known status. Unknown status cannot be combined with confirmed absence merely to complete a two-by-two table.

In another invented illustration, 192 participants are eligible, 180 have known status, and 36 of these have the event. The observed-status proportion is $36/180 = 20\%$. Dividing 36 by 192 gives 18.75%, but that calculation treats the 12 unknown statuses as contributing no recorded events to the full denominator.

Without additional information, between 36 and 48 of the 192 could have the event if every unknown status is respectively absent or present. The corresponding 18.75% to 25% range is a simple accounting bound. It is not a confidence interval and does not resolve the missingness mechanism.

Report the known-status calculation with its denominator and missing count. If a full eligible-population quantity is intended, justify how unknown outcomes are handled. The design also matters: case-control sampling does not generally allow the sampled event fraction to be interpreted as population risk.

For a binary model, verify which category is the event and which predictor level is the reference. The labels 0 and 1 need an explicit dictionary. A change in coding can reverse an odds-ratio comparison without changing the underlying records.

6.10 Prepare assumption checks without automatic test switching

Inspect appropriate summaries and graphs for the chosen question. Independent group comparisons need group distributions and counts. Paired inference needs within-pair differences.

Regression needs the relationship and relevant residual patterns. Categorical procedures need the observed table and, for an approximation, attention to expected counts.

A preliminary test result should not operate as an automatic switch between methods that target different quantities. A significant normality test does not by itself establish that a mean is no longer the research target. A non-significant test does not prove that the relevant model is adequate.

Resolve data errors first. Then evaluate whether the method is suitable, whether an alternative preserves or changes the target quantity, and whether the available sample supports the intended inference. Document that reasoning in the analysis record.

6.11 Keep the analytical record reproducible

Save the reviewed source version, dictionary, cleaning log, derivation definitions, eligibility rules, required-variable list, method settings, and analytical sample counts. The saved jamovi analysis should be identifiable as belonging to that record, with software and module versions documented when relevant.

A CSV export can carry values without carrying the full set of analysis settings and transformation definitions. Retain the saved analysis file as well as a suitable data export when both are needed for review. Confirm what survives each export rather than assuming that every format preserves the same information.

Reopen the saved file and verify that the intended variables, filters, counts, and results remain available. If an updated source is imported, repeat the count and derivation checks. A refreshed file may produce different results even when the analytical options have not changed.

6.12 Report the final sample and decisions

Describe the eligible population, required measurements, missing-data handling, and method. Give group counts or complete-pair counts where relevant. Explain exclusions using their actual reasons rather than a single unexplained phrase such as “incomplete data were removed.”

For the complete-pair teaching example, a defensible methods statement is: “The paired analysis required both baseline and follow-up measurements. Of 120 eligible participants, 108 had a complete pair; six lacked baseline, eight lacked follow-up, and two lacked both. Change was defined as follow-up minus baseline.”

The result should then use those 108 pairs and acknowledge what incomplete follow-up means for the interpretation. Do not borrow an estimate from a different sample or the separate 30-pair illustration to complete the paragraph.

6.13 Hands-on

> jamovi workspace

6.14 Review questions

1. Why can two valid analyses of the same dataset use different sample sizes?
2. How many complete pairs remain when 6 of 120 lack baseline, 8 lack follow-up, and 2 lack both?
3. Why is the event accounting bound different from a confidence interval?
4. What should be checked when a saved analysis unexpectedly has fewer participants?

6.15 Answers and reasoning

1. They can require different variables, eligibility rules, matching, or missing-data settings. Reconcile each count with its specification.
2. There are 108: subtract the union of missing measurements, 6 + 8 - 2 = 12, from 120.
3. It assigns extreme possible statuses to unknown outcomes. It does not quantify sampling uncertainty or justify a missingness model.
4. Check active filters, eligibility, required-variable completeness, coding, pairing, and the source version before attributing the change to the statistical procedure.

Part II

Descriptive Statistics and Graphs

Description answers what was observed: how many participants were analysed, how variables were distributed, which values were missing, and how groups differed in their observed characteristics. A descriptive table should therefore identify the denominator and measurement unit, not merely display numbers.

The next two chapters address numerical summaries and graphical presentation. Use both. A mean and standard deviation can conceal skewness, multiple peaks, or an influential observation. A graph can reveal those features but may not communicate the exact count or estimate needed for reporting. The NIST handbook describes these complementary numerical and graphical approaches to exploratory analysis. [20, 21]

Describe the data before comparing groups. This order helps identify coding problems and clarifies which statistical quantity would answer the question. A choice between a mean and a median should reflect the objective and distribution, rather than a rule that every variable must be reduced to the same summary.

By the end of this part, the reader should be able to construct a descriptive table, select a suitable graph, and explain the important features of an observed distribution.

Chapter 7

Descriptive Statistics

7.1 Learning objectives

After this chapter, the reader should be able to report counts with denominators, select appropriate measures of central value and spread, distinguish standard deviation from standard error, and interpret a descriptive estimate with its uncertainty.

7.2 Describe the observed sample first

Start with the number of eligible observations, the number included in the analysis, and the availability of key variables. Then describe variables using summaries that retain their meaning and units. Description is a check on the data and a basis for later method selection.

Suppose 200 participants are enrolled and 180 provide a follow-up measurement. The enrolled sample and the observed follow-up sample are different denominators. A report should state which denominator is used for each summary, rather than placing 200 at the top of every table column.

7.3 Frequencies, proportions, and percentages

A frequency is a count. A proportion divides a count by a defined denominator. A percentage multiplies a proportion by 100. In an invented example, 40 participants with an event among 200 participants with known event status gives

$$\hat{p} = \frac{40}{200} = 0.20 = 20\%.$$

Write "40 of 200 participants (20.0%)" rather than "20%" alone. The count and denominator indicate how much information supports the proportion. The same percentage based on 5 observations and 5,000 observations has very different estimation precision.

Distinguish percentages of all eligible participants from percentages among those with an observed value. If 40 events are recorded among 180 participants with known status and 20 statuses are unknown, the observed proportion is 40/180, or 22.2%. Reporting 40/200 would require an explicit assumption about the unknown statuses; it is not a neutral alternative.

7.3.1 Row and column percentages

In a table comparing event status by group, a row percentage can answer "What fraction of group A had an event?" A column percentage can answer "What fraction of participants with an event

belonged to group A?" These are different conditional descriptions. State the denominator in the table title or note.

jamovi's contingency-table analysis can display observed counts and row, column, or total percentages. Select the percentage that answers the question and check the orientation of the table. [22]

7.4 Mean, median, and mode

The arithmetic mean is the sum of values divided by their number. The median is the middle ordered value, using the mean of the two middle values when the sample size is even. The mode identifies the most frequent value or category. These summaries describe different aspects of a distribution. [23]

Consider invented waiting times, in minutes:

$$2,\ 3,\ 3,\ 4,\ 4,\ 5,\ 6,\ 7,\ 8,\ 18.$$

The sum is 60, so the mean is 6.0 minutes. The middle values are 4 and 5, so the median is 4.5 minutes. The long wait of 18 minutes pulls the mean upward. Neither result is a calculation error; they answer different descriptive questions about the same observations.

If the concern is a typical person's waiting experience, the median and spread of the central values may be helpful. If the concern is total waiting time divided by the number of attendees, the mean has a direct role. Select the summary according to the question and show enough of the distribution to avoid concealing the long waits.

7.5 Standard deviation, variance, range, and IQR

The sample standard deviation describes dispersion around the mean and retains the original measurement unit. Variance is its square. The range spans the minimum and maximum. The interquartile range is the 75th percentile minus the 25th percentile and describes the width of the middle half of the distribution. [21]

For the waiting-time example, the sample SD is 4.62 minutes. The minimum is 2 and the maximum is 18 minutes. Using the interpolation convention in the accompanying verification script, the first and third quartiles are 3.25 and 6.75 minutes, giving an IQR of 3.50 minutes.

Quartile algorithms can give different values in a small sample; report the method when exact reproduction matters. R documents several quantile definitions, with type 7 used by default in its `quantile()` function. [24]

When writing a median summary, give the two quartile limits rather than an ambiguous single "IQR" number if readers need to see the central interval: "median 4.5 minutes, first–third quartile 3.25–6.75." For a mean summary, write "mean 6.0 minutes (SD 4.6)." Preserve the distinction between a measure of location and a measure of spread.

7.6 Standard deviation is not standard error

The SD describes variation among individual observations. The SE describes the sampling variability of an estimate. For a mean of independent observations, the estimated SE is

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}.$$

The SD and sample size determine the standard error used in the conventional confidence interval for a mean. [25]

In an invented sample with SD 18 mmHg and 100 observations, the SE of the mean is 1.8 mmHg. This does not mean that individual blood pressures differ from the mean by only 1.8 mmHg. The smaller number describes uncertainty about the estimated mean, not the variation among people.

Increasing sample size can reduce the SE while the observed individual variation remains similar. When the purpose is to describe participants, report the SD. When the purpose is to estimate a population mean, report the mean with an appropriate confidence interval.

7.7 Confidence intervals for descriptive estimates

A confidence interval adds an uncertainty summary to a point estimate. For a mean, a conventional interval uses the estimated SE and a t-distribution critical value under its assumptions. For a proportion, several interval methods exist, and the method should be identified rather than assumed from the percentage alone. An exact binomial interval is one documented option. [26]

For an invented mean of 136 mmHg, SD 18 mmHg, and 100 independent observations, the conventional two-sided 95% t interval is 132.43 to 139.57 mmHg. This example concerns uncertainty in the mean. It is not an interval describing where 95% of individual blood pressure values lie.

Interpret confidence as a property of the interval-generating procedure under its assumptions. In repeated sampling, a 95% procedure covers the population parameter in approximately 95% of repetitions. It is not a frequentist probability of 95% that the fixed parameter lies within this particular realised interval. The next chapter develops this interpretation in more detail. [25]

7.8 Choose summaries according to the variable and question

Situation	Useful summary	What else should be shown
Nominal or binary variable	Count and percentage	Denominator, missing values, and category definition
Ordinal category	Counts in the defined order	Category labels and missing values
Approximately symmetric quantitative distribution	Mean and SD	Units, sample size, and a graph
Skewed distribution or notable long tail	Median and quartile limits; mean when relevant to the estimand	Range or graph showing the tail
Count outcome with many zeros	Counts or distributional summary suited to the objective	Frequency of zero, range, and possible overdispersion
Population estimate	Point estimate with CI	Interval method and sampling/design assumptions

These are starting choices, not automatic rules. A skewed variable can still have a mean as the target quantity. An ordinal scale may have a validated scoring convention that needs to be described. Explain why a summary answers the particular question.

jamovi's Descriptives analysis provides counts, missingness, location and spread summaries, and graphical options. Select the required quantities deliberately rather than transferring the entire output into a manuscript. [27]

7.9 Describe groups without overinterpreting them

Group-specific summaries show the observed distributions and denominators. They do not alone establish whether an intervention caused a difference or whether a population contrast is estimated precisely. Those judgments require the design and inferential method.

For an illustrative table, display the group label, number with an observed value, mean with SD or median with quartile limits, and measurement unit. Use the same summary convention across groups unless a difference is explained. Describe missing values separately instead of embedding them in the denominator silently.

7.10 Reporting example

For the invented waiting times: "Waiting time was available for 10 observations. The median was 4.5 minutes (first–third quartile 3.25–6.75), with values ranging from 2 to 18 minutes. The mean was 6.0 minutes (SD 4.6), reflecting the influence of the longest wait."

This statement gives the availability, location, spread, unit, and reason for displaying both mean and median. A report would usually choose the summaries most relevant to its objective rather than repeat all available numbers.

7.11 Common errors

Avoid giving percentages without denominators, using SEM to describe individual variation, averaging category codes, reporting identical denominators despite variable-specific missingness, or treating a confidence interval for the mean as the range of individual measurements. Check that each number has the intended role before interpreting it.

7.12 Hands-on

> jamovi workspace

7.13 Review questions

1. In the waiting-time example, why are the mean and median different?
2. With SD 18 and n = 100, what does SE = 1.8 describe?
3. If 20 of 200 event statuses are unknown, what denominator is used for the proportion among known statuses?

7.14 Answers and reasoning

1. The high value of 18 increases the arithmetic mean more than it changes the middle ordered values.
2. It estimates sampling variability of the mean under the independent-observation calculation, not variation among individual observations.
3. The denominator is 180. State the 20 unknown statuses separately and explain any alternative analytical assumption.

Chapter 8

Graphical Presentation

8.1 Learning objectives

After this chapter, the reader should be able to select a graph for a stated question, distinguish a categorical bar chart from a histogram, interpret distributional and relationship plots, and label uncertainty without confusing it with individual variation.

8.2 Let the question determine the graph

A graph should make one analytical task easier. A bar chart compares counts or proportions across categories. A histogram describes the distribution of a quantitative measurement. A scatterplot examines the relationship between two quantitative variables. A point with an interval displays an estimate and its uncertainty.

Before creating a graph, state what the reader should be able to see. "Compare the event proportions across groups" is a clearer task than "make a chart of the dataset." The first task determines the denominator, category labels, and axis scale; the second leaves those decisions unresolved.

Analytical question	Useful graph	Essential context
How many observations are in each category?	Bar chart	Counts, category definitions, and total
How is a quantitative variable distributed?	Histogram or distribution plot	Measurement unit, binning, and available n
How do distributions differ across groups?	Boxplot with individual points when feasible	Group n, median/quartile definition, and units
How do paired values change?	Paired-point plot with connecting lines	Participant matching and time order
Do two measurements vary together?	Scatterplot	Both units, observation unit, and relevant groups
How precise is an estimate?	Point and interval plot	Estimate, interval meaning, confidence level, and reference value

This table is a design guide for the manuscript. It should be applied to the specific question rather than used as an automatic conversion from variable type to picture.

8.3 Bar charts for categories

Use separate bars for categories and label the axis as a count or percentage. Absolute count bars should start at zero so that their lengths correspond to the values. If percentages are shown, state whether they use the total sample, a group denominator, or only observations with known status.

For nominal categories, ordering may follow a meaningful convention or the magnitude of the count. For ordinal categories, preserve the order of the scale. Sorting mild, moderate, and severe by frequency would make the display harder to interpret as an ordered severity distribution.

Avoid three-dimensional effects and decorative shapes that make lengths difficult to compare. A pie chart can show a coarse part-to-whole division with a few exclusive categories, but a bar chart usually makes precise comparisons easier. If category membership overlaps, the displayed percentages should not be presented as parts of one exclusive whole.

8.4 Histograms for numerical distributions

A histogram divides a numerical axis into intervals and counts observations within them. It can show location, spread, asymmetry, multiple peaks, and potential unusual observations. The width of the intervals affects the apparent shape, so interpretation should not depend on one arbitrary choice of bins. [20]

Unlike a categorical bar chart, neighbouring histogram bins represent adjacent numerical intervals. The horizontal axis therefore retains a quantitative scale. Label the vertical axis accurately: frequency, relative frequency, or density are different representations and should not be used interchangeably.

The invented waiting times from the descriptive chapter illustrate a long upper tail. Most values are between 2 and 8 minutes, with one value of 18 minutes. The histogram shows where values are concentrated; the boxplot shows the central interval and the separated high observation.

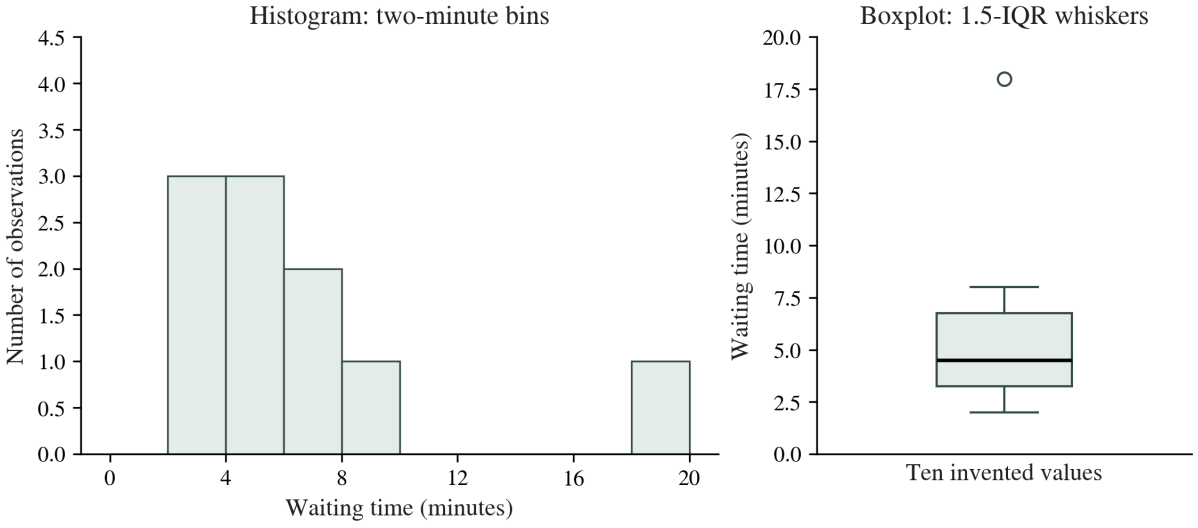


Figure 8.1: Histogram and boxplot of the ten invented waiting times. Both panels use the same measurements; they emphasise different distributional features.

A second peak may represent a mixture of groups, a change in measurement procedures, or a feature of the underlying distribution. Inspect relevant grouping and timing variables before treating the pooled distribution as one homogeneous sample.

8.5 Boxplots and individual points

A boxplot usually displays the median and first–third quartile interval. A common fenced version uses whiskers extending to the most extreme observations within 1.5 IQR of the quartiles and displays observations beyond those limits separately. The whisker convention must be understood because it can differ across implementations. [28]

An observation beyond a whisker is flagged by a plotting rule; the graph has not established that it is erroneous. Verify the source and examine its relevance to the question. A valid high value may be an important part of the distribution.

For small groups, displaying individual observations alongside a boxplot can make the amount of data visible. A boxplot alone may hide whether a group has five observations or fifty. Show the group sample sizes and use a consistent scale across panels when comparing magnitudes.

8.6 Q–Q plots and distributional assumptions

A normal Q–Q plot compares the ordered observed values with values expected under a normal reference distribution. Approximate alignment with a straight line supports a normal approximation; systematic bends or tail departures identify features needing further assessment. In a regression, the relevant plot is typically for residuals rather than for every raw predictor. [29]

A Q–Q plot does not replace the design assumptions. Perfectly aligned observations can still come from repeated measurements treated incorrectly as independent. Conversely, a small tail departure does not by itself identify the best alternative analysis. Interpret the plot alongside sample size, unusual observations, and the target quantity.

8.7 Scatterplots before correlation

Plot one point per defined observation unit with the two variables on numerical axes. Inspect direction, form, spread, unusual points, and possible subgroups. A straight upward pattern suggests a positive linear relationship. A steadily increasing curve is monotonic but not necessarily linear. A U-shaped pattern can contain a strong relationship even when its linear correlation is close to zero.

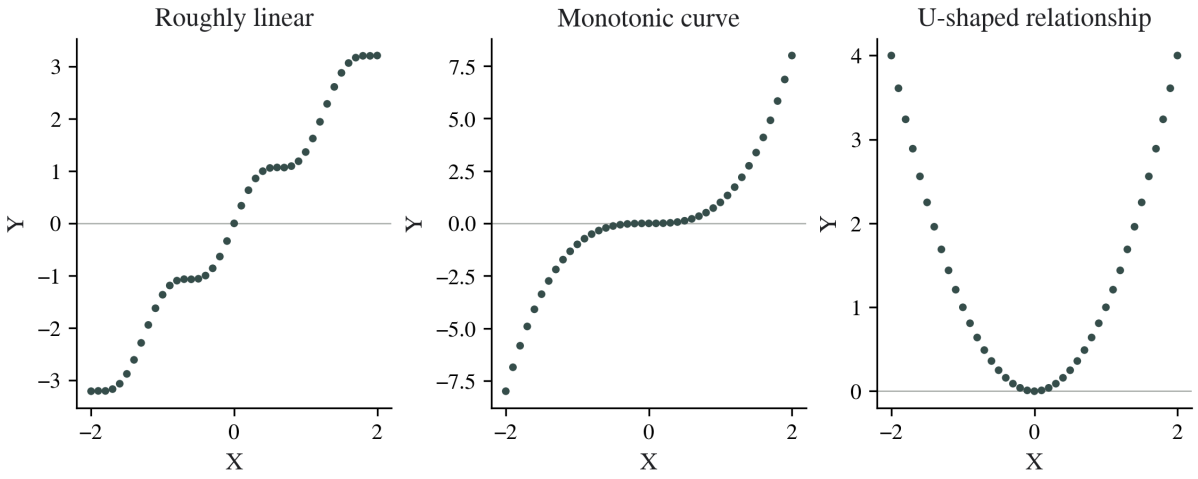


Figure 8.2: Three invented coordinate examples distinguish a roughly linear relationship, a monotonic curve, and a U-shaped relationship. The axes are arbitrary numerical units; panels use separate vertical scales to compare their form.

Do not draw connecting lines between unrelated individuals. Lines imply an order or a relationship between adjacent points. Use a line only when the connection represents something defined, such as repeated measurements from one participant or an ordered time series.

The correlation chapter develops the distinction between Pearson's linear coefficient and Spearman's rank-based coefficient. The scatterplot provides the context in which those numbers should be read. [30]

8.8 Paired measurements and time

For two measurements from the same participant, connecting that participant's two values preserves the matching. A plot of change values can then show whether changes are consistently in one direction or whether a few participants account for most of the mean change.

If follow-up times differ, state the time definition and window. A line from baseline to a common follow-up label should not conceal that observations were taken at substantially different times. More extensive longitudinal patterns require a design and analysis that account for their repeated structure.

8.9 Display uncertainty explicitly

An interval around a mean may describe a confidence interval, one standard deviation, a standard error, or a prediction interval. These have different meanings. State the interval type and level in the caption; a label of "error bars" alone is insufficient.

A CI for a mean describes estimation uncertainty. An SD describes individual spread. A prediction interval concerns a new observation under its model. A narrow CI therefore does not imply that individual values are tightly clustered. Similarly, overlap between two separate group intervals is not a direct test of the difference between those groups.

When the research question concerns a contrast, display the contrast itself with its interval and a relevant reference value. For a mean difference, the no-difference value is zero; for a ratio, the null value is one. Explain the chosen direction and scale in the caption.

8.10 Build a self-contained caption

A useful caption identifies the population or illustrative context, outcome and unit, groups or time points, sample size, and meaning of plotted summaries. If a graph shows invented teaching values, label them as invented. If a scale is transformed, state the transformation.

For example: "Distribution of invented waiting times for 10 observations. The histogram uses two-minute bins. The box shows the median and first–third quartiles; whiskers follow the 1.5-IQR convention. Points beyond the whiskers are plotted individually."

The ICMJE guidance describes figures as part of the scientific report and requires units and legends that allow their meaning to be assessed. Apply that principle to a teaching figure as well as to a manuscript figure. [31]

8.11 Common errors

Avoid missing denominators, unexplained axis truncation, different units across comparable panels, decorative three-dimensional bars, lines joining unrelated observations, and uncertainty intervals without a definition. Check the graph at its final printed size; a readable analysis window can become unreadable when reduced to a book page.

8.12 Hands-on

jamovi workspace

8.13 Review questions

1. What distinguishes a histogram from a categorical bar chart?
2. Does a point outside a boxplot whisker establish a data-entry error?
3. Why should a graph caption define what its intervals represent?

8.14 Answers and reasoning

1. Histogram bins represent adjacent numerical intervals; categories in a bar chart are discrete labels or levels.
2. No. The point is flagged by the plotting convention and needs source verification and analytical assessment.
3. CI, SD, SE, and prediction intervals answer different questions. The reader cannot infer the intended meaning from an unlabelled bar.

Part III

Core Inferential Methods

Inference uses observed data to learn about a quantity beyond the sample. Its validity depends on the study design and analysis assumptions. Increasing sample size can reduce sampling uncertainty, but it does not by itself remove a systematic error in how participants were selected or measurements were obtained.

This part begins with estimates, confidence intervals, p-values, errors, and power. It then considers continuous outcomes in two or more groups and associations between categorical variables. Throughout, define the target quantity before selecting the test: a mean difference, a distributional comparison, a proportion, or a ratio.

The p-value and confidence interval provide different information. The p-value refers to the incompatibility of the observed data with a specified null model. The interval communicates estimation uncertainty under its assumptions. Neither replaces the magnitude of the estimate or the clinical question. [1, 25]

By the end of this part, the reader should be able to choose a basic independent or paired comparison, check the relevant assumptions, interpret uncertainty, and avoid treating a non-significant result as proof that two groups are equivalent.

Chapter 9

Estimation and Hypothesis Testing

9.1 Learning objectives

After this chapter, the reader should be able to define a null hypothesis on the correct scale; explain a p-value and confidence interval; distinguish statistical from clinical importance; interpret Type I and Type II errors; and recognise why a non-significant result does not establish equivalence.

9.2 Begin with an estimate

Suppose the question is whether mean follow-up blood pressure differs between groups A and B. The target quantity is the population mean in A minus the population mean in B. The sample provides an estimate of that quantity. A statistical analysis should report the estimate and its uncertainty even when it also conducts a hypothesis test.

The null value depends on the scale. A difference of zero represents no difference; a ratio of one represents equal quantities. Testing an odds ratio against zero would therefore be a scale error. Specify the estimand, group direction, and null value before interpreting the output.

9.3 Sampling distributions and the normal model

A sampling distribution describes how an estimator varies across repeated samples. It is different from the distribution of individual measurements in one sample. A normal distribution is symmetric and is described by a mean and standard deviation; the standard normal has mean zero and SD one. [32]

Under appropriate conditions, including independent, identically distributed sampling and finite positive variance in the basic version, the central limit theorem makes the distribution of a properly standardised sample mean approach a normal distribution as sample size increases. This does not require the individual measurements themselves to become normal. There is no universal sample-size cutoff that removes all problems from heavy tails, dependence, or influential observations. [33]

The t-based intervals introduced here account for estimating the unknown variance under the conventional model. Always connect the reference distribution to the statistic and assumptions rather than treating “normality” as a label applied to an entire dataset.

9.4 Null and alternative hypotheses

The null hypothesis is the specified value or relationship being evaluated by a test. For a two-group mean comparison,

$$H_0 : \mu_A - \mu_B = 0.$$

A two-sided alternative is

$$H_1 : \mu_A - \mu_B \neq 0.$$

The model includes more than this equation: it also contains assumptions about the observations and their distribution. A p-value must be interpreted in relation to that model, not as a free-standing measure of truth.

A one-sided alternative addresses a prespecified direction. It should be justified by the question before the results are examined. Switching to a one-sided test after observing the direction of a difference changes the analysis procedure and its error properties.

9.5 What a p-value means

A p-value is the probability, under the specified null model, of a test statistic at least as incompatible with that model as the observed statistic, according to the chosen test. It is not the probability that the null hypothesis is true, the probability that the results arose "by chance," or the magnitude of the effect. [1]

The conditioning is essential: the calculation begins by assuming the null model and evaluating the observed statistic within it. It does not assign a probability to the model itself. A small value identifies incompatibility with the null model but does not identify which assumption or feature accounts for that incompatibility.

Read the p-value alongside the estimate, CI, sample size, design, and analysis decisions. An extremely small difference can produce a small p-value in a large sample. A substantial estimate can have a large p-value when the data provide little precision. Neither situation is resolved by the threshold alone.

9.6 Confidence intervals and sampling uncertainty

A confidence interval is produced by a procedure intended to cover a population parameter at a stated rate under its assumptions. In repeated sampling, a 95% procedure covers the parameter in approximately 95% of repetitions. Once one interval is observed, the fixed parameter is either inside or outside it; the confidence level is not a frequentist probability assigned to that particular inclusion. [25]

The interval can be read as a range of parameter values compatible with the observed data and model according to the interval procedure. This interpretation remains conditional on the design and analysis assumptions. It does not incorporate every possible bias or guarantee that values just outside the limits are impossible.

The figure demonstrates coverage across repetitions. It does not show 30 plausible positions for the same observed interval, and it is not a clinical dataset. Individual simulations need not yield exactly 95% coverage in a finite set of repetitions.

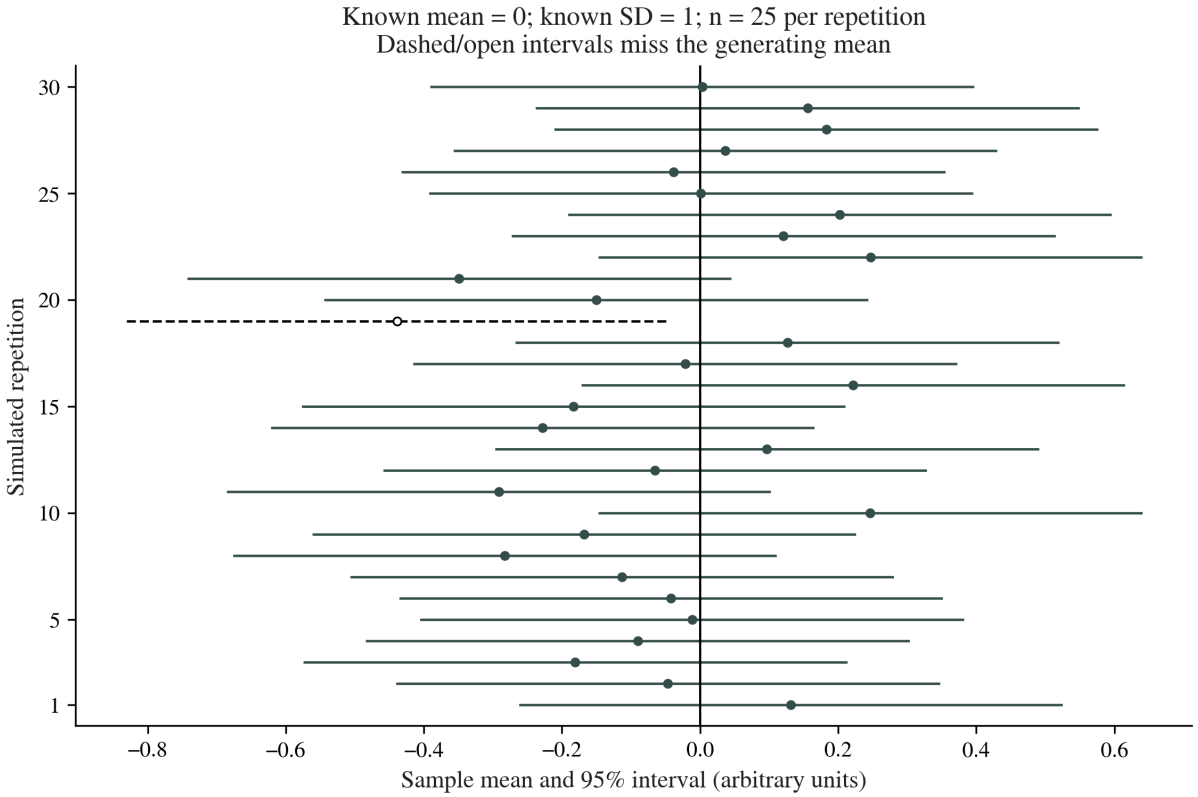


Figure 9.1: Repeated-sampling illustration of confidence intervals for a known mean. Each line is calculated from an invented simulated sample, and the vertical reference is the generating mean.

9.7 Precision and clinical importance

Precision concerns the uncertainty of an estimate. Clinical importance concerns whether its magnitude matters for the stated clinical or research objective. Establish the relevant threshold or interpretation using subject knowledge and appropriate sources before drawing a conclusion. A teaching threshold should be labelled as illustrative.

Consider three invented results for the same mean contrast:

Estimate and 95% CI, mmHg	What the result shows	What it does not establish
−6.0, CI −11.4 to −0.6	Negative estimate with an interval excluding zero	A precisely established reduction of at least an illustrative 5 mmHg
−6.0, CI −13.0 to 1.0	Negative estimate with substantial uncertainty, including zero	Absence of a clinically relevant difference
−1.0, CI −2.0 to −0.1	Small negative estimate with an interval excluding zero	Clinical importance merely because zero is excluded

The second and third intervals are schematic teaching examples, not calculated study results. The first is reproduced in the two-group chapter. All three require a stated clinical question before being described as useful or important.

9.8 Type I and Type II errors

A Type I error is rejection of a true null hypothesis under the test procedure. The significance level, alpha, sets its nominal control under the relevant assumptions. A Type II error is failure to reject the null when a specified alternative is true. Power is one minus the Type II error probability for that alternative. [34]

Power is not a property of a sample size alone. It depends on the specified effect, variability, analysis, significance level, and other design features. A study can have good power for a large effect and poor power for a smaller effect. Define the effect that matters before calculating the required sample size. R's power calculation documentation makes these inputs explicit for t-tests. [35]

Do not interpret alpha as the probability that a particular positive result is false. That statement conditions on a different event and requires additional information. Similarly, planned power is not the probability that one non-significant result is a false negative. The probability definitions concern repetitions of a specified procedure and model.

9.9 Non-significance and equivalence

When a test does not reject the null, the data may be compatible with zero and with a range of non-zero values. The CI indicates how broad that range is. A statement of "no difference" is especially misleading when the interval includes differences that would matter for the study objective. [36]

Equivalence and non-inferiority require their own prespecified margins, hypotheses, and analyses. A conventional superiority test with p greater than 0.05 does not answer those questions. In this volume, report the estimated difference and interval, then explain whether the remaining uncertainty includes relevant effects.

9.10 Multiple analyses and selective interpretation

If many outcomes or subgroup comparisons are tested, the opportunity to obtain a small p-value somewhere increases. Define the primary question and distinguish prespecified analyses from exploration. Where multiple comparisons belong to one inferential family, select and report a suitable adjustment strategy. [37]

Adjusting p-values does not repair selective reporting of only the most favourable analysis. Report the relevant analysis family and explain which conclusions are confirmatory and which are exploratory. A change in method after seeing the result should be documented with its reason.

9.11 Assumptions belong to the interpretation

Every test is attached to an observation structure and a statistical model. Independence, pairing, category definitions, and measurement units should be checked before examining distributional assumptions. A normality check cannot establish that repeated observations are independent.

A diagnostic test's non-significant result does not prove that its null hypothesis is true. Use plots, design information, and the sensitivity of the intended analysis to departures, rather than choosing methods solely by whether a preliminary p-value falls above or below 0.05. [6]

9.12 Reporting an inferential result

A clear statement identifies the comparison and analytical sample, gives the estimate with units, reports the CI and method, and includes the test result when it is relevant. Its interpretation should remain proportionate to the design and uncertainty.

For the calculated teaching contrast: "Mean follow-up systolic blood pressure was 6.0 mmHg lower in group A than in group B (A minus B: −6.0 mmHg; 95% CI −11.4 to −0.6; Welch's two-sided $p = 0.031$). The interval allows effects smaller and larger than the illustrative 5-mmHg threshold, so the estimate does not establish that the reduction is at least 5 mmHg."

9.13 Hands-on

jamovi workspace

9.14 Review questions

1. Why is $p = 0.03$ not the probability that the null hypothesis is true?
2. A mean difference is −6 with CI −13 to 1. Does this establish no difference?
3. What is the no-association value for an odds ratio?
4. Why is power incomplete without a specified alternative effect?

9.15 Answers and reasoning

1. The p-value is calculated conditional on the null model and concerns the test statistic, not the probability of the model.
2. No. The interval includes zero and meaningful negative values; the estimate remains uncertain.
3. One. Equal odds give a ratio of one.
4. The chance of rejecting the null changes with the alternative effect. One design can detect a large effect more readily than a smaller one.

Chapter 10

Comparing Two Groups or Two Paired Measurements

10.1 Learning objectives

After this chapter, the reader should be able to distinguish independent from paired comparisons; define the contrast direction; choose between a mean-based and rank-based question; check the relevant assumptions; and report an estimate without reducing the analysis to a p-value.

10.2 Establish the comparison structure

Two groups are independent when each observation contributes to one group and the sampling or design does not create the paired dependence addressed by a paired method. Two measurements are paired when they belong to the same participant or to a defined matched pair. The distinction comes from the design, not from whether the group sample sizes happen to be equal.

Examples are mean follow-up blood pressure in separate groups A and B, and baseline versus follow-up blood pressure in the same participants. The first asks about a between-group contrast. The second asks about the mean within-participant change. These questions should not be analysed interchangeably.

Feature	Independent comparison	Paired comparison
Basic unit	One observation in one group	One complete pair
Main mean contrast	Mean in A minus mean in B	Mean of defined within-pair differences
Dependence handled	Groups are independent under the simple model	Matching is retained within pairs
Key graphical checks	Group distributions and unusual observations	Distribution of within-pair differences
Missingness	Available outcome values in each group	Availability of both measurements for the paired calculation

10.3 Decide what should be compared

A comparison of means targets an arithmetic difference. A rank-based comparison asks about the ordering or distribution of observations under its assumptions. A non-parametric test is not simply a mean test with the normality requirement removed.

Specify which quantity matters. If the objective concerns average resource use, a mean may remain relevant even when the distribution is skewed. If it concerns an ordered outcome or a distributional shift, a rank-based approach may be suitable. The choice should retain the original research objective.

10.4 Independent-samples t-tests

10.4.1 Welch's method

Welch's t-test estimates a mean difference using the two sample variances separately. It does not impose the equal-variance assumption of the pooled Student method. R documents the Welch/Satterthwaite degrees-of-freedom calculation and distinguishes it from the pooled-variance option. [38]

For groups A and B,

$$\widehat{\Delta} = \bar{x}_A - \bar{x}_B, \qquad SE(\widehat{\Delta}) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}.$$

The test compares this difference with the specified null value using the appropriate t reference distribution. A confidence interval adds and subtracts the relevant critical value multiplied by the SE.

Welch's method addresses unequal variances; it does not repair dependence, invalid coding, or an outcome dominated by a few influential observations. Inspect those issues separately. In small samples, major departures from the intended distributional model require particular attention.

10.4.2 Pooled Student's method

The pooled independent t-test uses a shared variance estimate. Its interpretation requires a defensible equal-variance model in addition to the independent-observation and distributional conditions. A preliminary variance test alone should not determine whether those assumptions are scientifically reasonable.

If the pooled method is used, explain why the common-variance model is appropriate. If Welch's method is used, report it by name rather than labelling every independent comparison simply as "a t-test." The choice affects the SE and degrees of freedom when group variances or sizes differ.

10.4.3 Check the data before reading the test

Confirm the group labels, outcome unit, group counts, missingness, and contrast direction. Inspect each group's distribution and check verified extreme values for their influence. Distinguish a coding error from a genuine unusual observation. The analysis should follow the specified estimand rather than change because one version gives a smaller p-value.

10.5 Calculated independent-group illustration

The following summaries are invented for teaching and are not extracted from a patient dataset:

Group	n	Mean systolic blood pressure, mmHg	SD, mmHg
A	60	128.0	14.0
B	60	134.0	16.0

The contrast A minus B is −6.0 mmHg. Its Welch SE is 2.745 mmHg. The t statistic is −2.186, with approximately 115.96 degrees of freedom. The two-sided p-value is 0.0308 and the 95% CI is −11.44 to −0.56 mmHg. These quantities are reproduced in the project's verification script.

The estimate indicates a lower mean in A. The interval excludes zero but still allows a wide range of reductions. If 5 mmHg is used as an explicitly illustrative importance threshold, the interval does not establish that the reduction is at least that large. The study design, rather than the t-test alone, determines whether the difference can be interpreted causally.

10.5.1 Reading the output

Locate the row for the chosen procedure and confirm the sign of the mean difference. Read the effect estimate and CI before the p-value. The degrees of freedom may be non-integer for Welch's procedure; this is a property of its approximation, not a sign that participants were counted fractionally.

jamovi's independent-samples analysis assigns a continuous outcome and a two-level grouping variable. The author will add the dataset, options, and output screenshots in the Hands-on section. [39]

10.5.2 A reasoning record for the illustration

The summary calculations can be reviewed now, while the practical demonstration remains reserved for the author. A complete reasoning record connects the numerical result with its intended question and its limits:

Stage	Decision or finding
Question	Compare mean follow-up systolic blood pressure in groups A and B
Target	The unadjusted mean difference, A minus B, in mmHg
Structure	One participant contributes to one group; this independence is an illustration assumption
Preparation	Verify group labels, outcome unit, eligibility, missingness, and one observation per participant
Procedure	Welch's two-sided comparison of means, with a 95% CI
Numerical finding	Estimated difference -6.0 mmHg; interval -11.44 to -0.56 mmHg
Meaning	A lower estimated mean in A, with uncertainty about the magnitude
Reporting	Include group summaries, contrast direction, CI, method, and design limitations

The two summary rows are sufficient for the stated Welch calculation. They are insufficient to inspect distributions, detect influential values, establish the missingness pattern, verify the observation unit, or establish a causal interpretation. Those checks require the study record and individual observations. Do not label them as completed merely because the calculation can be reproduced.

If the scientific objective is a baseline-adjusted group contrast, this unadjusted result does not answer the full question. If observations are clustered or repeated, the assumed independent structure is inadequate. These are reasons to revise the analytical method while preserving the stated objective, rather than simply choose another test from the same menu.

Reversing the contrast to B minus A gives +6.0 mmHg and a 95% CI of +0.56 to +11.44 mmHg. The two-sided evidence is unchanged, but the reporting direction must change with it. This is a useful consistency check when reviewing the future jamovi output.

10.6 Paired t-test

The paired t-test analyses the within-pair differences. Define the direction, such as follow-up minus baseline, calculate a difference for each complete pair, and estimate the mean difference with its SE. The distributional assumption used for the conventional small-sample t inference concerns the differences, not separate normality tests for the two measurement columns. [40]

For differences d_i,

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad SE(\bar{d}) = \frac{s_d}{\sqrt{n}}.$$

The model requires independence between the pairs for this simple analysis. It does not require independence between the two measurements within each pair; that dependence is why the pairing is retained.

Confirm matching by identifier rather than row order alone when measurements were imported from separate files. Excluding a participant with one missing measurement changes the complete-pair sample. Report how many pairs are available and examine whether incomplete follow-up limits the interpretation.

10.6.1 Calculated paired illustration

In an invented example of 30 complete pairs, the mean baseline value is 146 mmHg and the mean follow-up value is 138 mmHg. The mean difference, follow-up minus baseline, is −8.0 mmHg, and the SD of the differences is 10.0 mmHg.

The paired SE is $10/\sqrt{30} = 1.826$ mmHg. The t statistic is −4.382 with 29 degrees of freedom. The 95% CI for mean change is −11.73 to −4.27 mmHg, and the two-sided p-value is 0.000141. Report $p < 0.001$ at an appropriate precision.

The estimate describes average within-participant change. A before–after study without an appropriate comparison does not isolate the reason for that change. Time trends, additional care, regression to the mean, and incomplete follow-up may remain relevant. The statistical result should not be labelled a treatment effect solely because the change is non-zero.

jamovi's paired analysis requires the defined measurement pairs and can provide mean-difference, interval, descriptive, and diagnostic outputs. Check the sign convention used in its displayed comparison against the convention specified in the manuscript. [41]

10.7 Mann–Whitney U test

The Mann–Whitney test uses the relative ranks of observations in two independent groups. It is useful for an ordered or distributional comparison under its assumptions. An interpretation as a simple location or median shift requires additional conditions, such as comparable distributional shapes; it is not automatically a test of two medians. [42]

For a small invented illustration, let A contain 2, 4, 6, and 8, and B contain 1, 3, 5, and 7. The combined ranks for A are 2, 4, 6, and 8, with rank sum 20. Subtracting $4(4+1)/2 = 10$ gives one convention for $U_A = 10$.

Among the 16 A–B pairs, A is higher in 10, so the observed probability-of-superiority summary is $10/16 = 0.625$, with no ties. This calculation helps explain the ordering information. It does not replace a test or interval appropriate to the chosen estimand, and the sample is too small for a precise population conclusion.

Software may report a rank-sum statistic or a transformed U statistic. Identify the definition when reproducing output. Ties and exact versus asymptotic calculations can also affect the reported p-value, so record the selected approach.

10.8 Wilcoxon signed-rank test

The signed-rank test is for paired differences or a one-sample difference distribution. It ranks the non-zero absolute differences and uses their signs. Its conventional location interpretation assumes a suitably symmetric difference distribution; it is not an assumption-free alternative for any paired dataset. [42]

For invented differences -4, -2, -1, 3, and 6, the ranks of the absolute differences are 4, 2, 1, 3, and 5. The positive rank sum is $3 + 5 = 8$, and the negative rank sum is $4 + 2 + 1 = 7$. This example shows why the signed-rank method uses both direction and ranked magnitude, rather than just counting positive changes.

If a difference distribution is strongly asymmetric, consider whether a different estimand or method is required. The paired t-test targets a mean change; the signed-rank procedure targets a different distributional feature. Changing tests should not be described as preserving the same question automatically.

10.9 Effect size and reporting

Retain the effect on its original scale when that scale is clinically interpretable. A -6-mmHg difference communicates more directly than a standardised label alone. If a standardised effect is added, define its denominator and sign convention. Paired and independent standardisations are not interchangeable.

A reporting sentence for the independent illustration is: "Group A had a lower mean follow-up systolic blood pressure than group B (128.0 versus 134.0 mmHg; n = 60 in each group). The mean difference, A minus B, was -6.0 mmHg (95% CI -11.4 to -0.6; Welch t = -2.19, df = 115.96, p = 0.031)."

For the paired illustration: "Among 30 complete pairs, mean follow-up minus baseline systolic blood pressure was -8.0 mmHg (SD of differences 10.0; 95% CI -11.7 to -4.3; paired t = -4.38, df = 29, $p < 0.001$)." The interpretation should then state the design limitations relevant to attributing the change.

10.10 Hands-on

jamovi workspace

10.11 Review questions

1. Two groups have the same number of observations. Does this make them paired?
2. Which distribution is relevant to the paired t-test's conventional normality assumption?
3. What does Welch's procedure address, and what does it leave unresolved?
4. Why is a Mann–Whitney result not automatically evidence of different medians?

10.12 Answers and reasoning

1. No. Pairing requires a defined relationship between observations.
2. The within-pair differences.
3. It relaxes the pooled equal-variance assumption. It does not resolve dependence, biased sampling, invalid coding, or every form of influential non-normality.
4. The procedure uses ranks and distributional ordering. A pure median or location-shift interpretation needs additional distributional conditions.

Chapter 11

Comparing More Than Two Groups

11.1 Learning objectives

After this chapter, the reader should be able to distinguish an overall comparison from a specific contrast, interpret a one-way ANOVA, recognise when unequal variances require a different method, and select compatible follow-up comparisons for mean-based or rank-based analyses.

11.2 Define groups, outcomes, and independence

Consider mean follow-up blood pressure across three independent groups. The outcome is continuous and the grouping variable has three levels. A one-way analysis considers one grouping factor. The number of groups does not change the need to define eligibility, units, missingness, and the observation structure.

Three measurements from the same participant are not three independent groups. Similarly, participants within shared facilities can have a clustered structure. These designs require an analysis that recognises their dependence, rather than an independent one-way ANOVA applied to every row.

Identify whether the objective is an overall assessment of group means or a particular comparison, such as intervention A versus a control. An overall test and a specific contrast answer different questions. Define the contrast when it is the primary scientific objective.

11.3 What a one-way ANOVA tests

The conventional one-way ANOVA evaluates the null hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k.$$

Its alternative is that the means are not all equal. A small omnibus p-value does not mean that every pair differs. It also does not identify the size or direction of any particular contrast.

The fixed-effects one-way model expresses an observation as a group-specific mean component plus an error. The conventional model assumes independent errors with a common variance and a normal distribution for the relevant inference. [43]

11.4 Why the procedure is called analysis of variance

ANOVA compares variation between group means with variation within groups. If the observed groups differ substantially relative to the variation among observations within groups, the resulting F statistic becomes larger under the comparison with the null model.

The total squared variation is divided into between-group and within-group components. Each is converted to a mean square by dividing by its degrees of freedom. For k independent groups and N observations,

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}, \qquad df_1 = k - 1, \quad df_2 = N - k.$$

The F value is a test statistic, not a measurement of a clinical difference. Retain group summaries and the relevant contrast estimates when reporting the result.

11.5 Assess the conventional assumptions

Independence is established by design and data structure. A plot cannot prove it. Inspect whether participants contribute once to the comparison and whether matching or clustering needs to be accounted for.

Distributional assessment concerns within-group deviations or model residuals, rather than the pooled outcome distribution alone. Three groups with different means can create a non-normal-looking pooled distribution even when the within-group model is reasonable. Examine the appropriate residual or group plots.

The conventional model also assumes a common variance. Consider the observed spreads, sample sizes, and sensitivity to departures. A non-significant preliminary variance test does not prove that the variances are equal. Choosing a method solely from that preliminary p-value is an incomplete decision rule.

11.6 Welch's one-way ANOVA

Welch's one-way procedure compares group means while relaxing the common-variance assumption. It uses group-specific variability and an adjusted reference distribution. R's `oneway.test()` documentation distinguishes the Welch approach from the equal-variance method, and jamovi's simplified One-Way ANOVA offers a Welch option. [13, 44]

Welch's method does not resolve dependence or a poor match between the intended estimand and the observed outcome. A skewed distribution with a few highly influential values still requires assessment. Report the selected method and its numerator and denominator degrees of freedom as displayed, including non-integer values where appropriate.

Choose follow-up procedures consistent with the variance model. A method based on a pooled common variance should not be added automatically after selecting an unequal-variance analysis.

11.7 Calculated three-group illustration

The following summaries are invented for teaching:

Group	n	Mean, mmHg	SD, mmHg
A	20	125.0	10.0
B	20	130.0	10.0

Group	n	Mean, mmHg	SD, mmHg
C	20	135.0	10.0

The overall mean is 130.0 mmHg. The between-group sum of squares is 1,000; the within-group sum of squares is 5,700. Dividing by 2 and 57 degrees of freedom gives mean squares of 500 and 100. The conventional F statistic is therefore 5.00, with df = 2 and 57, and p = 0.00998.

The analysis indicates incompatibility with the equal-means model under its assumptions. It does not establish that A differs from B, B differs from C, and A differs from C individually. Those contrasts require their own estimates and inferential procedure.

The sample eta-squared summary is

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} = \frac{1000}{6700} = 0.149.$$

This describes the between-group share of the observed squared variation in this example. It is not the percentage of participants who improved or a causal percentage explained by treatment. Prefer original-scale contrast estimates when interpreting the health question.

11.8 Planned contrasts and post-hoc comparisons

A planned contrast is specified from the research objective before inspecting the results. For example, a study may compare the average of two intervention groups with a control. Its coefficients and uncertainty calculation should match that defined comparison.

Post-hoc pairwise comparisons assess particular pairs, often after an overall exploration. With three groups there are three distinct pairs; with four there are six. State which family of comparisons is being evaluated and which adjustment or simultaneous interval procedure is used.

Tukey's procedure provides simultaneous intervals for pairwise mean differences under its model. Games–Howell is an unequal-variance pairwise approach available in jamovi's One-Way ANOVA documentation. These methods serve different assumptions and should be selected consistently with the main analysis. [13, 45]

A scientifically planned contrast need not be treated as a question that exists only after a significant omnibus test. Explain the inferential plan and multiplicity control appropriate to that objective. Conversely, reporting only the favourable pair after many unreported comparisons is selective reporting.

11.9 Multiple comparisons and error control

An adjustment applies to a defined family of analyses. Bonferroni controls a familywise error rate by a simple correction; Holm is a sequential familywise procedure. False-discovery-rate methods address a different error criterion. The intended criterion and family should be stated. [37]

Do not describe an adjusted p-value without identifying what was adjusted. A correction across the three group pairs is different from one across several outcomes, subgroup comparisons, and time points. The analysis plan should define the relevant family rather than allow it to change to obtain a preferred result.

An adjustment also does not turn an exploratory analysis into a prespecified confirmatory one. Preserve that distinction in the interpretation.

11.10 Kruskal–Wallis test

The Kruskal–Wallis procedure uses ranks for comparisons across independent groups. It is appropriate for a defined rank-based or distributional question under its assumptions. Its location or median interpretation requires additional conditions about distributional shapes; it should not automatically be described as an ANOVA of medians. [46]

As with ANOVA, a global result does not identify which groups differ. Follow-up rank comparisons require a compatible method and a declared multiplicity approach. R documents pairwise Wilcoxon comparisons with p-value adjustment as one implementation route; the procedure chosen in jamovi should be identified by its actual name and version. [47, 48]

Inspect group distributions before using a pure location interpretation. If one group differs mainly in spread and another in location, a global rank result may reflect more than a difference in central values. Show medians and quartile limits as descriptions without claiming that these alone define the test's null hypothesis.

11.11 Reading jamovi output

Confirm the continuous outcome and grouping variable, then identify the conventional, Welch, or non-parametric procedure actually requested. Read group counts, summaries, diagnostic information, the overall statistic, and relevant contrast output. Check how missing observations were excluded. [13, 48]

When pairwise tables are present, verify the reference order of each difference, the CI or adjustment method, and the family to which the comparisons belong. Different contrast directions can coexist in one output table; a sign should never be interpreted without its labels.

11.12 Reporting example

For the calculated conventional illustration: "The invented mean follow-up values were 125.0, 130.0, and 135.0 mmHg in groups A, B, and C, respectively (20 observations per group; SD 10.0 in each group). A conventional one-way ANOVA gave $F(2, 57) = 5.00$, $p = 0.010$. The overall comparison does not identify the individual group contrasts; these would require the prespecified follow-up analysis."

For a real report, replace the final teaching statement with the relevant contrast estimates and intervals. State why the selected variance model and observation structure are appropriate, and identify the follow-up method if one is used.

11.13 Common errors

Avoid applying independent ANOVA to repeated measurements, describing the omnibus result as evidence that all pairs differ, choosing post-hoc methods that contradict the variance model, treating Kruskal–Wallis as an unconditional median test, or reporting pairwise p-values without the adjustment strategy.

11.14 Hands-on

jamovi workspace

11.15 Review questions

1. What does the alternative hypothesis of a one-way ANOVA mean?
2. Why are three follow-up measurements from the same participant not three independent groups?
3. What must be identified when reporting an adjusted pairwise p-value?
4. Does a Kruskal–Wallis result always establish different medians?

11.16 Answers and reasoning

1. The means are not all equal; it does not assert that every pair is different.
2. Measurements are linked within participants and require an analysis that retains that dependence.
3. State the procedure, the family of comparisons, and the direction of the contrast.
4. No. The method uses ranks, and a median-shift interpretation requires additional distributional conditions.

Chapter 12

Associations Between Categorical Variables

12.1 Learning objectives

After this chapter, the reader should be able to construct a contingency table, choose the correct denominator, distinguish independent from paired categorical observations, assess expected counts, and interpret a difference or ratio on the scale supported by the study design.

12.2 Start with the design and denominator

Consider whether an event occurs more often in group A than in group B. For an independent comparison, each participant contributes one group assignment and one event status. The table describes two categorical variables observed on the same set of independent participants.

Before calculating a percentage, establish who was eligible to contribute to the event denominator and whether status is known. A recorded absence is different from an unknown outcome. If follow-up is incomplete, a simple event proportion may fail to answer the intended risk question.

The study design determines the interpretation. An event proportion over defined follow-up can describe risk when its denominator and ascertainment support that meaning. A cross-sectional proportion may describe prevalence. A rate uses an appropriate person-time denominator and is a different quantity. In a case-control sample, selection by outcome means the sampled case proportion is not the population risk.

12.3 Construct and label the contingency table

The following independent-group example is invented for teaching. Assume complete event status over the same defined follow-up in both groups:

Group	Event present	Event absent	Total
A	30	70	100
B	15	85	100
Total	45	155	200

The group-specific event proportions are $30/100 = 30\%$ and $15/100 = 15\%$. These are row percentages in this orientation. The fraction of all event cases belonging to A is $30/45 = 66.7\%$, a different denominator and question.

State the reference direction. The calculations below compare A with B. Reversing that direction changes the sign of a difference and takes the reciprocal of a ratio.

12.4 Pearson chi-square test of independence

The null model states that the two categorical variables are independent. It specifies expected cell counts calculated from the row and column totals:

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{N}.$$

The test statistic summarises the observed deviations from these expected counts:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad df = (r-1)(c-1).$$

This is a test of association, not a measure of the magnitude of the association. R's documentation describes the Pearson procedure, continuity correction for a 2-by-2 table, and a simulation-based option. [49]

In the invented table, the expected event count in each group is 22.5, and the expected absent count is 77.5. The uncorrected Pearson statistic is 6.452 with one degree of freedom, giving p = 0.0111. The estimate of association still needs to be reported separately.

12.5 Expected counts and sparse tables

The adequacy of the chi-square approximation concerns **expected** counts, not simply observed counts or overall sample size. A large total sample can still contain sparse cells when one category is rare. Examine the table dimensions and the distribution of expected counts.

One commonly used screening convention checks for no expected count below one and a high proportion of expected counts at least five. R's documented hybrid Fisher procedure uses more than 80% at least five and none below one. These are approximation rules within a defined procedure, not a universal guarantee of validity for every table. [50]

For a sparse 2-by-2 table, Fisher's exact procedure is a common option. For larger tables, an appropriate exact or Monte Carlo procedure may be needed, subject to computational feasibility and software availability. State the method and, for simulations, the number of repetitions and relevant reproducibility settings.

Yates continuity correction applies to a 2-by-2 chi-square calculation. It is not a general repair for sparse tables of arbitrary dimensions. Select the procedure using observation structure, dimensions, expected counts, and the question rather than overall sample size alone. [49]

Do not combine categories merely to make a test run unless the grouping is substantively defensible. Category merging changes the question and should be documented.

12.6 Fisher's exact procedure

Fisher's test evaluates the table conditional on its margins under the independence model. For the invented 2-by-2 table, the two-sided p-value is 0.0171. This differs from the uncorrected Pearson p-value because the procedures use different inferential calculations.

An "exact" calculation does not remove the need for appropriate observation structure or eliminate every uncertainty about design. The term identifies how the probability is calculated under

the model. The two-sided definition and reported odds-ratio estimator can also vary by procedure; R's Fisher documentation uses a conditional estimate rather than the simple cross-product ratio. [50]

When reporting, identify the chosen procedure rather than selecting whichever p-value is smaller. The point estimate and interval should also be labelled with their calculation method.

12.7 Risk or prevalence difference

Under the stated teaching follow-up, the risk difference is

$$RD = \frac{30}{100} - \frac{15}{100} = 0.15.$$

This is **15 percentage points**, not a 15% relative increase. Its large-sample Wald 95% CI in this illustration is approximately 3.6 to 26.4 percentage points. The absolute difference communicates how much the group-specific proportions differ on the original probability scale.

If the same table arose from a cross-sectional study, use an interpretation appropriate to prevalence rather than follow-up risk. If it arose from outcome-based sampling, do not calculate a population risk difference from the sampled case fractions.

12.8 Risk ratio and odds ratio

For the teaching follow-up table,

$$RR = \frac{30/100}{15/100} = 2.00.$$

The group A event risk is twice that in B under the stipulated design and complete follow-up. A log-scale large-sample 95% CI is 1.15 to 3.48.

Odds are probability divided by its complement. Group A odds are 30/70; group B odds are 15/85. The sample odds ratio is

$$OR = \frac{30/70}{15/85} = \frac{30 \times 85}{70 \times 15} = 2.43.$$

Its log-Wald 95% CI is 1.21 to 4.87. The odds ratio and risk ratio differ because odds and probability differ. An OR of 2.43 here is not a statement that the event probability is 2.43 times as high. Bland and Altman's Statistics Note explains the odds-ratio calculation and interpretation. [51]

The approximation of an OR to an RR depends on the outcome probabilities and context; it should not be assumed without checking. Always name the reported measure and reference category.

12.8.1 Interpret interval methods carefully

The Wald intervals above are teaching calculations appropriate to the stated positive cell counts and large-sample approximation. They are not presented as the exact intervals returned by Fisher's procedure. With small or zero counts, the simple log formulas can fail or perform poorly, requiring a suitable alternative method.

Measure in the example	Estimate	Interpretation of A relative to B
Risk difference	0.15	15 percentage points higher risk
Risk ratio	2.00	Twice the risk
Odds ratio	2.43	2.43 times the odds

The numerical direction is the same, but the magnitudes are not interchangeable. A reporting table should not use one generic heading such as "relative risk" for all three quantities.

12.9 Larger contingency tables

For more than two rows or columns, the overall chi-square test evaluates independence across the complete table. Its result does not identify a particular category as the cause of the association. Examine observed counts, relevant percentages, and prespecified comparisons before drawing a category-specific conclusion.

Cramér's V can summarise the strength of an association without giving its clinical direction. It is not a risk ratio, an odds ratio, or the proportion of participants with an event. Prefer question-specific contrasts when available and substantively useful.

jamovi's contingency-table analysis can display observed and expected counts, percentages, test results, and several association measures. Select the quantities appropriate to the table and design; an available checkbox does not establish that every measure is meaningful for that study. [22]

12.10 Paired binary outcomes: McNemar's question

If the same participants have a binary status before and after an observation period, the data are paired. A 2-by-2 table now cross-classifies before and after status rather than two independent groups. McNemar's procedure evaluates marginal symmetry and uses the discordant pairs for the comparison. [52]

In an invented table, suppose 20 participants have the event at both times, 15 change from present to absent, 5 change from absent to present, and 60 are absent at both times. There are 100 pairs, but only the 20 discordant pairs inform the change-direction comparison. The before proportion is 35% and the after proportion is 25%.

An independent Pearson test applied as if the before and after samples were unrelated would discard the pairing. Use a paired categorical procedure, with an exact conditional approach considered when the discordant counts require it. A binomial calculation on the discordant counts is one exact route under the paired null model. [26, 53]

12.11 Reporting example

For the independent teaching illustration: "The event occurred in 30 of 100 participants in group A (30.0%) and 15 of 100 in B (15.0%). Under the stipulated common follow-up, the risk ratio for A versus B was 2.00 (log-Wald 95% CI 1.15–3.48), and the absolute risk difference was 15.0 percentage points (Wald 95% CI 3.6–26.4). The uncorrected Pearson chi-square test gave $^2(1) = 6.45$, $p = 0.011$."

This statement names the denominators, measures, directions, interval methods, and test. It does not describe the association as causal. If a different design or procedure is used, revise those elements rather than transplanting this wording unchanged.

12.12 Hands-on

jamovi workspace

12.13 Review questions

1. Which counts determine whether the chi-square approximation may be inadequate?
2. Does a large total sample guarantee that all table cells have adequate expected counts?
3. In the invented table, why are RR = 2.00 and OR = 2.43 different?
4. Which observations inform McNemar's comparison in the paired example?

12.14 Answers and reasoning

1. Expected counts under the independence model, assessed with table dimensions and the distribution of sparse cells.
2. No. A rare category can create sparse expected counts despite a large total.
3. The RR compares probabilities; the OR compares probability divided by its complement.
4. The 15 present-to-absent and 5 absent-to-present pairs. The concordant pairs do not contribute to that direction comparison.

Part IV

Correlation and Introductory Regression

Association asks whether two variables vary together. Regression goes further by specifying an outcome and describing how its expected value changes with a predictor. These methods require an explicit data structure and a clear interpretation of the analysis scale.

The correlation chapter distinguishes linear and monotonic relationships and explains why a scatterplot is needed before interpreting a coefficient. The linear regression chapter introduces a continuous outcome with one predictor, the meaning of a slope, and basic residual assessment. The logistic chapter introduces a binary outcome, odds, and the transformation between model coefficients and odds ratios.

The emphasis in this volume is interpretation of simple, unadjusted relationships. Multiple predictors, confounder selection, interaction, detailed model strategy, and predictive validation are reserved for later volumes. A fitted association should not be described as an independent or causal effect merely because it is obtained from a regression model.

jamovi's regression documentation provides the corresponding output quantities and available analysis options. The text develops the interpretation of those quantities; the author will add the practical demonstrations in the Hands-on boxes. [54–56]

Chapter 13

Correlation

13.1 Learning objectives

After this chapter, the reader should be able to distinguish linear from monotonic association, interpret direction and magnitude without using universal strength labels, identify the relevant data pairs, and explain why correlation is neither causation nor agreement.

13.2 Define the association question

Consider whether age and systolic blood pressure vary together among study participants. Each participant contributes one age–blood-pressure pair. Neither variable is automatically an outcome in the correlation calculation; the purpose is to summarise their relationship.

Confirm that paired values belong to the same observation unit. If a participant contributes multiple visits, a conventional correlation across all rows does not account for the dependence between those visits. A correlation between facility averages also answers a different question from a correlation between individual measurements.

State whether the objective concerns a linear relationship in the original values or an ordered relationship. The choice between Pearson and Spearman should reflect that objective and the observed form, not simply a threshold from a normality test.

13.3 Inspect the scatterplot first

Look for direction, form, spread, influential observations, and subgroups. The graphical-presentation chapter shows examples of a roughly linear relationship, a monotonic curve, and a U-shaped relationship. A single coefficient cannot communicate all those features.

A pooled association can depend on differences between subgroups rather than on the relationship within them. Inspect groups relevant to the question before treating a pooled coefficient as the whole explanation. This inspection does not itself adjust for confounding; it reveals a feature that may require a different analysis.

When a verified unusual observation strongly affects the coefficient, show or describe that influence. Do not remove it solely to obtain a preferred value. If a sensitivity analysis is justified, report the rule and its interpretation.

13.4 Pearson's correlation coefficient

Pearson's r describes linear association after scaling the covariance by the two standard deviations:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}.$$

It ranges from -1 to $+1$ when both variables have non-zero variation. A positive value indicates that larger values tend to occur together in a linear pattern; a negative value indicates the opposite direction. A constant variable does not provide the variation needed for this coefficient.

The coefficient is a descriptive calculation and does not require normality merely to exist. Its conventional small-sample test and interval use additional model assumptions. R documents the t-based Pearson test under independent observations from a suitable normal model and a Fisher-transformation interval. [30]

13.4.1 Direction and magnitude

The sign communicates direction. The absolute magnitude indicates how closely the observed values follow a linear relationship in their standardised scale. It does not indicate the slope in mmHg per year or the size of a clinical effect in original units.

Avoid using "weak," "moderate," or "strong" as universal clinical conclusions. Such labels are conventions, and their value depends on the measurement reliability, population range, and research objective. Report the numerical coefficient, uncertainty, sample size, and scatterplot context before adding any descriptive label.

13.4.2 Why zero does not mean no relationship

A near-zero Pearson coefficient can occur when positive and negative linear components balance in a curved pattern. In a symmetric U-shaped example, outcomes vary systematically with the predictor while the linear coefficient is zero. The graph is therefore necessary to distinguish lack of a linear association from lack of any relationship.

13.5 Spearman's rank correlation

Spearman's coefficient applies correlation to the ranks of the paired values. It describes monotonic association: whether values tend to move in one direction together, without requiring a straight-line pattern in the original units. Ties need an appropriate ranking and inferential calculation. [30]

An increasing curve can have a high rank correlation even when its original-scale relationship is not linear. A U-shaped relationship is not generally monotonic over its full range, so substituting Spearman for Pearson does not automatically summarise every curve well.

Choose Spearman because ranks and monotonic ordering match the question and data. Do not describe it as a method that ignores all assumptions or repairs invalid observation matching. Independence between pairs remains relevant to the conventional inferential calculation.

13.6 Inference for Pearson's coefficient

For a conventional test of zero population correlation,

$$t = r\sqrt{\frac{n-2}{1-r^2}}, \qquad df = n-2.$$

The p-value refers to that null model. It is not the probability that the correlation is causal or that a future sample will show the same magnitude.

An approximate interval can be formed using Fisher's transformation:

$$z = \operatorname{atanh}(r), \qquad SE(z) = \frac{1}{\sqrt{n-3}},$$

then transforming the interval limits back with the hyperbolic tangent. This is an interval for Pearson's population correlation under the method's assumptions. Do not use it as an unlabelled interval for a different coefficient.

13.7 Calculated illustration

Suppose an invented Pearson result is $r = 0.60$ from 50 complete, independent pairs. The t statistic is approximately 5.20 with 48 degrees of freedom, giving a two-sided p-value of 0.00000412. The approximate Fisher 95% CI is 0.386 to 0.753.

The estimate indicates positive linear association, but the interval permits a range of magnitudes. It does not say that a one-year age increase changes blood pressure by 0.60 mmHg. That original-scale question requires a regression slope or another suitable model.

For comparison, an invented $r = 0.30$ with the same 50 pairs has p = 0.0343 and a Fisher interval of approximately 0.024 to 0.534. A threshold label alone would hide the uncertainty in its magnitude. These are separate numerical teaching cases, not outcomes selected from one clinical dataset.

13.8 Correlation is not causation

An association may reflect a causal pathway, a common cause, selection, measurement structure, or other features of the study. A correlation coefficient does not distinguish these explanations. The design and substantive knowledge determine which interpretation is defensible.

For example, a relationship between age and blood pressure does not specify what would happen if one could intervene on age, nor does it isolate the influence of other characteristics. Later volumes address adjustment and causal reasoning. This volume reports the observed relationship with an explicit boundary on interpretation.

13.9 Correlation is not agreement

Two measurement methods can correlate perfectly while differing systematically. If method B equals method A plus 20 units for every observation, the linear correlation is +1, but the methods do not give the same values. A method-comparison question therefore requires assessment of differences and agreement, not correlation alone.

Reliability, agreement, and diagnostic accuracy have different targets. A high correlation does not establish that one method can replace another clinically. The relevant specialist methods are developed in later volumes.

13.10 Correlation matrices and missingness

A matrix can present correlations among several variables. Inspect which observations contribute to each pair and whether the sample sizes differ. A coefficient based on 50 complete pairs and another based on 35 pairs do not have the same information, even if they appear in one matrix.

jamovi's correlation analysis provides Pearson, Spearman, and Kendall options, sample-size display, and graphical and uncertainty options. Verify which outputs apply to the selected coefficient and record the method. [54]

If many correlations are examined, distinguish the prespecified relationship from exploratory screening. Multiplicity and selective interpretation remain relevant; a matrix of flagged p-values is not a model-selection strategy by itself.

13.11 The relationship with simple regression

In an ordinary least-squares regression with one quantitative predictor, an intercept, and the same complete observations, $R^2 = r^2$. This relationship does not make the two analyses interchangeable. Correlation is symmetric in the two variables, while regression specifies an outcome and produces a slope in defined units.

The regression chapter explains why those units, residuals, and prediction uncertainty matter. A numerical association summary should lead to a clearer question, not to an automatic causal explanation.

13.12 Reporting example

For the invented 50-pair result: "Age and systolic blood pressure showed a positive linear association (Pearson r = 0.60; approximate Fisher 95% CI 0.39–0.75; n = 50; $p < 0.001$). Interpretation requires the scatterplot and the observation structure; the coefficient does not establish a causal effect or an original-scale increase per year."

In a completed study report, describe the actual scatterplot assessment and any influential observations or design limitation. Avoid adding an unsupported strength category when the numerical result and context are more informative.

13.13 Hands-on

> jamovi workspace

13.14 Review questions

1. Can Pearson's r be calculated for non-normal measurements?
2. Does r close to zero rule out a U-shaped relationship?
3. Why can perfect correlation coexist with poor agreement?
4. What does r = 0.60 fail to tell the reader about the slope in original units?

13.15 Answers and reasoning

1. Yes, if the required variation and data pairs exist. The conventional test and interval require additional assumptions.
2. No. It summarises linear association, and opposite sides of the curve can balance.
3. A systematic offset or scale difference can preserve linear association while changing the measured values.
4. It does not provide the outcome change per predictor unit; that requires a model with the relevant units.

Chapter 14

Introduction to Simple Linear Regression

14.1 Learning objectives

After this chapter, the reader should be able to specify a continuous outcome and one predictor, interpret an original-scale slope and intercept, distinguish fitted values from residuals, recognise the assumptions for conventional inference, and separate estimation of a mean from prediction of an individual value.

14.2 Specify the outcome and predictor

Suppose the question is how mean systolic blood pressure varies with age in the observed study population. Blood pressure is the outcome, and age is the predictor. Each independent participant contributes an outcome–predictor pair.

Simple regression uses one predictor. It does not estimate an effect independent of all other patient characteristics. Multiple predictors, confounder selection, interaction, and adjusted interpretation are reserved for Volume II and beyond. Calling a coefficient "independent" merely because it appears in a regression table would overstate this model.

14.3 The simple linear model

The model expresses the outcome as

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad E(\varepsilon_i \mid X_i) = 0.$$

The intercept β_0 is the model's conditional mean when $X = 0$. The slope β_1 is the difference in conditional mean associated with a one-unit increase in X. The error represents the departure of an individual outcome from that conditional mean.

Ordinary least squares chooses coefficients that minimise the sum of squared residuals. R's `lm()` documentation identifies this as a route for fitting linear models and describes the model specification. [57]

The model is linear in its coefficients. The basic chapter uses a straight-line relationship with one predictor so that the slope's units and interpretation are clear. More flexible relationships are developed later rather than forced into the same interpretation.

14.4 Interpret the slope in its units

In a deliberately specified teaching equation,

$$\widehat{\text{SBP}} = 100 + 0.50 \times \text{age},$$

the slope is 0.50 mmHg per year. A ten-year age difference corresponds to a five-mmHg difference in the fitted conditional mean under this equation. It is an association in the specified model, not a claim that changing age causes blood pressure to change.

The fitted mean at age 40 is 120 mmHg; at age 50 it is 125 mmHg. These are values generated by the teaching equation, not predictions validated in a clinical population. Do not use them to guide care.

Scaling the predictor changes the numerical coefficient without changing the underlying fitted relationship. If age is expressed in decades, the same slope becomes 5.0 mmHg per decade. Always state the predictor unit.

14.5 Interpret the intercept only where meaningful

The teaching equation gives an intercept of 100 mmHg at age zero. If the study includes adults only, age zero is outside the observed range. The intercept is part of the mathematical equation but may not have a useful substantive interpretation in that study.

Centering age at 40 would write the same fitted relationship as

$$\widehat{\text{SBP}} = 120 + 0.50 \times (\text{age} - 40).$$

The intercept now describes the fitted mean at age 40, while the slope and fitted values are unchanged. This illustrates why a reference value can make an intercept easier to interpret.

14.6 Fitted values and residuals

A fitted value is the outcome value on the estimated mean line for a given predictor value. A residual is observed minus fitted:

$$e_i = y_i - \hat{y}_i.$$

If an invented participant aged 40 has observed SBP 132 mmHg, the teaching equation gives a fitted value of 120 and a residual of +12 mmHg. The residual indicates where that observation lies relative to the mean line; it is not a model coefficient.

Residual plots assess whether the mean relationship and error structure are suitable. Curvature, changing spread, and influential observations can identify aspects that a single fit summary conceals. NIST emphasises that a high R^2 does not establish adequate model fit. [58]

14.7 Assumptions and what they are needed for

The conditional mean should be represented adequately by the specified model. Observation or error dependence should be handled according to the study design. The conventional equal-variance SE assumes an appropriate constant error variance. Normality of errors supplies the conventional exact small-sample t inference, rather than a requirement that the raw predictor be normally distributed.

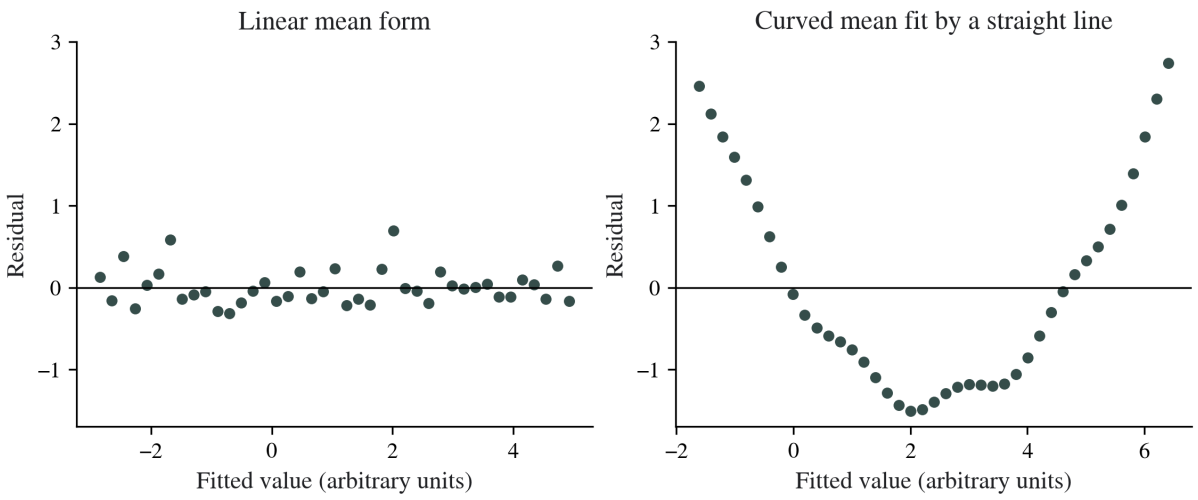


Figure 14.1: A correctly shaped linear teaching relationship has residuals without a systematic curve; fitting a straight line to a curved relationship leaves a visible residual pattern. All coordinates are invented for illustration.

These conditions have different roles. It is possible to calculate an ordinary least-squares line without establishing that its usual confidence interval is valid. Examine the estimation and inference assumptions separately instead of treating "the regression ran" as confirmation of both.

Check	What to inspect	Interpretation
Mean relationship	Outcome–predictor plot and residual pattern	A curve can suggest the straight-line mean is inadequate
Observation structure	Sampling, repeated records, and clusters	Dependence requires a suitable model or variance approach
Error spread	Residuals across fitted values	Changing spread can affect the usual SE calculation
Error distribution	Residual Q–Q plot	Relevant to conventional small-sample inference
Influence	Verified unusual predictor or outcome combinations	Assess effect on the fit; do not delete automatically

A high-leverage observation is unusual in its predictor position. A large residual is unusual relative to the fitted outcome. An influential observation changes the fitted model materially. These descriptions overlap but are not identical.

14.8 Confidence intervals and hypothesis tests for the slope

A conventional slope interval has the form

$$\hat{\beta}_1 \pm t_{1-\alpha/2,n-2}\, SE(\hat{\beta}_1),$$

under the simple model's assumptions. The corresponding null slope is zero. An interval excluding zero describes evidence of a non-zero linear association under that model; it does not establish a clinically important or causal effect.

For an invented summary with n = 60, slope 0.50 mmHg/year, and SE 0.10, the t statistic is 5.00 with 58 degrees of freedom. The 95% interval is approximately 0.30 to 0.70 mmHg/year.

This summary is constructed for teaching; it is separate from the correlation example's 50 observations.

Read the coefficient estimate and interval with its units. A standardised coefficient uses a different scale and should not be substituted silently for the original-scale slope.

14.9 What R-squared describes

With an intercept, R^2 describes the proportion of the observed outcome sum of squared deviations accounted for by the fitted model. It is not the percentage of people correctly predicted or a guarantee that the model is useful in a new setting.

For the invented simple-model summary above, the corresponding R^2 is $25/(25 + 58) = 0.301$. This algebra follows from the simple regression test relationship. The remaining outcome variation is not made irrelevant by a non-zero slope.

A high R^2 can coexist with a poor functional form, influential observations, or invalid prediction outside the data. Check residuals and the intended use rather than selecting a model solely because this number is large. [58]

14.10 Estimating a mean versus predicting an individual

A confidence interval for the fitted conditional mean addresses uncertainty about the mean at a specified predictor value. A prediction interval for a new individual also includes the unexplained individual variation. Under the model, the latter is wider because it answers a different question. [59]

Do not use a narrow mean interval to suggest that individual outcomes can be predicted equally precisely. Similarly, an estimated mean relationship within the observed age range does not validate extrapolation to an age far outside that range.

Prediction-model development requires additional assessment of performance and validation. Those tasks are beyond this introductory association model and are reserved for later volumes.

14.11 A binary predictor as a mean comparison

A predictor coded 0 for group B and 1 for A gives an intercept representing the mean in B and a slope representing A minus B in the simple unadjusted model. This links regression coefficients to the mean-difference question introduced earlier.

The usual pooled regression SE corresponds to its variance model. It should not be assumed to reproduce the unequal-variance Welch calculation merely because the coefficient estimate is the same. Preserve the intended inference assumptions as well as the estimand.

14.12 Reading jamovi output

Confirm the continuous outcome and single predictor. Identify the original-scale coefficient, SE, confidence interval, test result, and the analysis sample. Examine the fit and residual information appropriate to the model. jamovi's linear regression documentation describes these outputs and diagnostic options. [55]

An output can also offer multiple predictors. Their availability does not mean that this introductory analysis should add variables without a research-based plan. Model specification belongs to the question, design, and later-volume adjustment strategy.

14.13 Reporting example

For the constructed coefficient summary: “In the simple unadjusted linear model, each additional year of age was associated with a 0.50-mmHg higher mean systolic blood pressure (95% CI 0.30–0.70; n = 60; t = 5.00, df = 58). The result describes a linear association and does not isolate age from other patient characteristics.”

A completed report should identify the actual residual assessment, included range, exclusions, and other limitations. Do not claim that assumptions were satisfied unless those checks were performed and documented.

14.14 Hands-on

jamovi workspace

14.15 Review questions

1. What are the units of a slope when SBP is in mmHg and age is in years?
2. Does a large R-squared establish that the straight-line model fits adequately?
3. Why is a prediction interval for a new individual wider than a mean interval?
4. Does a one-predictor coefficient establish an independent effect after adjustment?

14.16 Answers and reasoning

1. mmHg per year.
2. No. The residual pattern and design may reveal inadequacy that the scalar fit summary conceals.
3. It includes both uncertainty in the mean relationship and individual outcome variation under the model.
4. No. A simple model contains one predictor and has not adjusted for other characteristics.

Chapter 15

Introduction to Logistic Regression

15.1 Learning objectives

After this chapter, the reader should be able to define a binary outcome and its event category, distinguish probability from odds, interpret a coefficient as an odds ratio, and recognise the limits of a simple unadjusted binary-outcome model.

15.2 The scope of this chapter

This is an introductory bridge from Volume I to Volume II. It explains the binary-outcome scale and a model with one predictor. Multiple predictors, confounder selection, interaction, detailed diagnostics, penalised estimation, and prediction-model validation belong to later volumes.

The chapter connects a contingency-table odds ratio with a regression coefficient. Its introductory role supports interpretation of the binary-outcome scale before moving to adjusted modeling.

15.3 A binary outcome needs a defined event

A binary outcome has two defined categories. Write which category is the event being modeled and which is the comparison. A model of event presence and a model of event absence describe complementary probabilities and reverse the odds-ratio direction.

For teaching, code event present as $Y = 1$ and event absent as $Y = 0$. The probability $p = P(Y = 1 \mid X)$ is the conditional event probability. In software, verify the actual event and reference labels in the output; do not infer them from a remembered default.

The observation unit and outcome window remain essential. One record per participant with a known event status is different from several visits per participant. A binary variable can be entered into the software in both situations, but the same simple independence model is not automatically appropriate.

15.4 Probability, odds, and log odds

Probability ranges from zero to one. Odds are

$$\text{odds} = \frac{p}{1-p}.$$

If the event probability is 0.15, the odds are $0.15/0.85 = 0.1765$. A probability of 0.30 gives odds of $0.30/0.70 = 0.4286$. The probability ratio is 2.00, while the odds ratio is 2.43. The measures describe different scales. [51]

The natural logarithm of the odds is the **logit**:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right).$$

Unlike probability, log odds can vary over the full numerical line. Logistic regression models that transformed quantity and maps it back to a probability between zero and one.

15.5 The simple logistic model

With one predictor,

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i.$$

The corresponding event probability is

$$p_i = \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)}.$$

The binomial family with a logit link is a documented generalised linear model specification. R's family and model documentation distinguish the response distribution from its link function. [60, 61]

A one-unit predictor increase adds β_1 to the log odds and multiplies the odds by $\exp(\beta_1)$. It does not add a fixed amount to the event probability. The probability change depends on the starting probability and the model.

15.6 A binary predictor connects the model to a table

Return to the invented independent-group table: 30 events among 100 participants in A and 15 among 100 in B. Let $X = 0$ for B and $X = 1$ for A.

The intercept is the log odds in B:

$$\beta_0 = \log(15/85) = -1.7346.$$

The group coefficient is the log odds ratio:

$$\beta_1 = \log\left(\frac{30/70}{15/85}\right) = 0.8873.$$

Exponentiating gives $\exp(0.8873) = 2.4286$. Transforming the linear predictor back to probability gives 0.15 for B and 0.30 for A. This reproduces the two observed teaching proportions in the simple two-group model.

The coefficient is unadjusted because the model contains only group. It should not be described as an association independent of age, severity, or other characteristics.

15.7 A continuous predictor and the probability curve

For a continuous predictor, the model assumes a straight-line relationship between that predictor and the log odds in this simple specification. It does not assume a straight-line relationship with the raw event probability. A one-unit effect on the odds can correspond to different absolute probability changes at different starting probabilities.

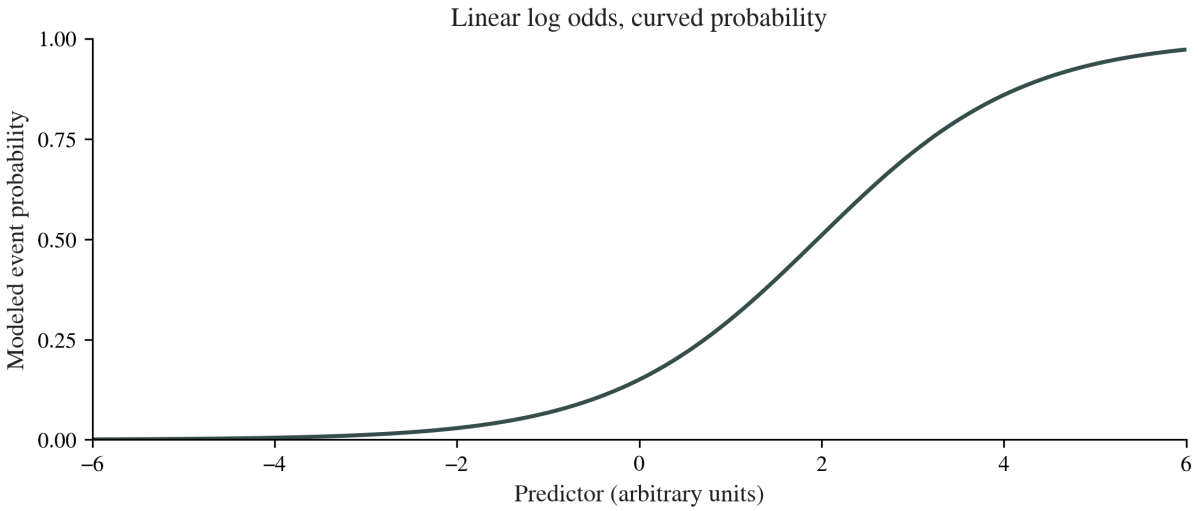


Figure 15.1: A teaching logistic curve maps a straight-line log-odds predictor to probabilities between zero and one. Its numerical predictor units are arbitrary and do not represent a validated clinical score.

State the predictor unit and consider a clinically understandable scaling before reporting a coefficient. A coefficient per year and one per decade differ numerically because the unit differs. Their odds ratios must be interpreted on the stated increment.

15.8 Confidence intervals and p-values

A common large-sample Wald interval is first calculated for the log-odds coefficient, then exponentiated to the OR scale. The no-association coefficient is zero; the corresponding OR is one. Other interval procedures may be appropriate, so identify the method rather than assuming all intervals are Wald intervals.

For the invented table, the group coefficient is 0.8873 with SE 0.3550. Its Wald z statistic is 2.499, p = 0.0124, and the exponentiated 95% interval is 1.21 to 4.87. These are calculated teaching values from the single-predictor table, not results of an author-supplied jamovi dataset.

The Wald p-value differs from the Pearson and Fisher p-values in the categorical chapter because these procedures use different test calculations. The relevant question is whether the selected procedure and its assumptions are appropriate, not which value is most favourable.

15.9 Basic model checks

Check event coding, predictor coding, reference category, analytical sample, and observation independence. For continuous predictors, assess whether the linear logit form is appropriate. Consider whether the data provide enough event and non-event information for the model being fitted.

Binary outcomes do not have normally distributed errors in the ordinary linear-model sense. A normality test of the 0/1 outcome is therefore not a prerequisite for logistic regression. The binomial variance changes with probability, and the model uses that response structure. [60]

Very sparse combinations or separation can produce unstable or diverging coefficient estimates. Separation occurs when predictor information divides event from non-event status perfectly in a way that prevents a finite ordinary maximum-likelihood estimate. Unusually large estimates, broad intervals, or fitting warnings require investigation rather than uncritical interpretation. [61]

Do not use a rigid event-count rule as proof that a model is adequate. The information, complexity, predictor distributions, and intended use matter. Detailed strategy for limited information and more complex models is reserved for Volume II and III.

15.10 Association and prediction are different objectives

An association analysis focuses on the predictor contrast and its interpretation. A prediction model focuses on estimating an individual's outcome probability in a specified setting. A significant OR does not by itself show that prediction is accurate or clinically useful.

Prediction requires assessment of calibration, discrimination, and validation appropriate to the use. A classification accuracy at one cutoff can be misleading when the event is uncommon, and the cutoff has to serve a specified decision. This chapter does not claim to validate a clinical prediction tool.

Pseudo-R^2 measures in logistic output are not interchangeable with the ordinary linear model's R^2. Do not label them as the literal percentage of individual outcome variation explained without identifying their definition.

15.11 Design affects interpretation of fitted probabilities

The teaching table stipulates complete common follow-up and supports its stated risk interpretation. In outcome-based sampling, the sampled event fraction does not directly reproduce the population event probability. An odds-ratio analysis may remain relevant, but absolute risks require additional design information and an appropriate method.

Likewise, a before–after binary outcome or matched case-control design requires recognition of pairing or matching. A simple unadjusted independent logistic model should not be transplanted into those structures without extension.

15.12 Reading jamovi output

Confirm the binary outcome, modeled event, reference level, and single predictor. Distinguish the coefficient on the log-odds scale from its exponentiated OR. Read the interval, SE or test quantity, analytical sample, and relevant model notes. jamovi's logistic documentation describes coefficient, odds-ratio, and model-fit outputs. [56]

The complete procedure, dataset, and screenshots will be added in the empty Hands-on box. This introductory text supports interpretation of that output; it does not claim that the missing practical demonstration has already been performed.

15.13 Reporting example

For the teaching table: “In a simple unadjusted logistic model with group B as reference, group A had higher event odds (OR 2.43; Wald 95% CI 1.21–4.87; p = 0.012). The model represents 200 invented participant outcomes, including 45 events. The OR compares odds, not risks, and no other predictors were included.”

For an actual manuscript, state the event definition and time window, reference category, selected model, missing-data handling, and design limitations. Avoid replacing “odds” with “risk” merely to simplify the sentence.

15.14 Hands-on

> jamovi workspace

15.15 Review questions

1. What is the OR when a coefficient on the log-odds scale is zero?
2. Does OR = 2 mean that event probability doubles?
3. What must be verified before interpreting the direction of a software coefficient?
4. Does the one-predictor teaching model adjust for age or other characteristics?

15.16 Answers and reasoning

1. $\exp(0) = 1$, the no-association value.
2. No. It means that odds double; the corresponding probability change depends on the initial probability.
3. Verify the modeled event, predictor coding, and reference category.
4. No. It contains group only and is unadjusted.

Part V

Reporting and Interpretation

A reader should be able to identify what was analysed, how the result was obtained, how large the estimate was, and how uncertain it remains. Reporting therefore starts with the question and method, followed by a result that retains the denominator, units, reference categories, and relevant uncertainty.

The reporting chapter brings together the preceding analyses. It explains the functions of a results paragraph, a table, a figure, and a methods statement. It also addresses common errors such as copying an entire software output, presenting only p-values, describing association as causation, and interpreting a non-significant result as absence of an effect.

The SAMPL guidelines provide a general basis for statistical reporting in biomedical manuscripts. Study-specific reporting guidance should also be selected according to the design; STROBE applies to the observational designs listed in its statement. [2, 3]

By the end of this part, the reader should be able to produce a concise results section that another researcher can interpret and assess without access to the jamovi output window.

Chapter 16

Reporting Statistical Results

16.1 Learning objectives

After this chapter, the reader should be able to write a methods statement and results paragraph for a basic analysis, design a self-contained table or figure, report uncertainty and missingness, and keep interpretation within the study's design and evidence.

16.2 Report the answer to the question

The results section should make the analytical finding visible. A software table contains many possible quantities; a manuscript selects those needed to answer the research question. Begin with the observation count and comparison, then present the estimate, uncertainty, and relevant inferential result.

The methods explain how the analysis was performed. The results state what it produced. The discussion interprets the result in relation to the question, limitations, and other evidence. Keeping these roles distinct reduces repetition and unsupported conclusions.

The SAMPL guidelines provide statistical reporting guidance, while ICMJE recommends sufficient methodological detail for a knowledgeable reader to evaluate and verify the analysis. Use the guideline appropriate to the study design alongside these general principles. [2, 31]

16.3 A minimum reporting structure

Element	What the reader needs
Population and sample	Who contributed and the analytical denominator
Outcome	Definition, unit, and measurement time
Comparison	Groups, reference category, and contrast direction
Method	Actual procedure, relevant settings, and interval method
Estimate	Magnitude on a named scale
Uncertainty	Interval and confidence level
Test information	Test statistic, degrees of freedom where relevant, and p-value
Limitations	Missingness, design, or model issues affecting interpretation

Not every paragraph needs every statistic. The reporting purpose determines which details belong in the main text, table, methods, or supplement. The essential information should nevertheless be recoverable from the report.

16.4 Write the methods before the results

Describe variable definitions, inclusion rules, summary conventions, and the intended analysis. Identify independent, paired, or repeated data. State how missing observations were handled and how uncertainty was calculated. Record software and module versions for the actual implementation.

A template for the independent mean illustration is: "Follow-up systolic blood pressure was compared between groups A and B using Welch's independent-samples t-test. The estimand was the mean in A minus the mean in B, expressed in mmHg. The mean difference and two-sided 95% CI were reported; the analytical denominator was the number with an observed follow-up value in each group."

This template does not assert that real diagnostic checks were completed. In an actual report, add the checks performed and how any important departures were addressed. Do not write "assumptions were satisfied" as a standard sentence copied across all analyses.

16.5 Report descriptive results

For a categorical variable, give the count and denominator with the percentage. For a quantitative variable, identify the summary and unit. Distinguish mean with SD from mean with CI; the first describes observed dispersion, while the second describes estimation uncertainty under its method.

For the invented waiting times: "The median waiting time was 4.5 minutes (first–third quartile 3.25–6.75; n = 10). Values ranged from 2 to 18 minutes." This is a descriptive statement. It does not need a hypothesis test simply to state what was observed.

If values are unavailable for some participants, report that availability explicitly. A table can include an observed n for each variable or a missing-value column. Choose a format that allows the denominator to be understood without excessive footnotes.

16.6 Report mean comparisons

For the invented independent groups: "Mean systolic blood pressure was 128.0 mmHg (SD 14.0) in A and 134.0 mmHg (SD 16.0) in B, with 60 observations in each group. The difference, A minus B, was −6.0 mmHg (95% CI −11.4 to −0.6; Welch t = −2.19, df = 115.96, p = 0.031)."

The sign and units make the result interpretable. "There was a significant difference" does not communicate the magnitude, direction, or precision. If the manuscript focuses on the contrast, the group summaries can sit in a table while the paragraph highlights the difference and CI.

For the paired example: "Among 30 complete pairs, mean follow-up minus baseline blood pressure was −8.0 mmHg (95% CI −11.7 to −4.3; paired t = −4.38, df = 29, p < 0.001)." State the complete-pair denominator and the direction of change.

16.7 Report categorical associations

Use the measure supported by the design and name it. A risk ratio, prevalence ratio, odds ratio, and absolute percentage-point difference are not interchangeable. Identify the event and reference group before reporting a ratio.

For the invented independent table: "Events occurred in 30/100 participants in A and 15/100 in B. Under the stipulated follow-up, the RR for A versus B was 2.00 (log-Wald 95% CI 1.15–3.48)." If an odds ratio is reported instead, write "OR 2.43 (log-Wald 95% CI 1.21–4.87)" and interpret it as odds.

Name the test and interval procedure. “Fisher’s exact p-value” and “Wald OR interval” describe different components; do not imply that they all came from one exact method unless that is what the procedure returned.

16.8 Report correlation and regression

For correlation, name the coefficient, provide its estimate, analytical pairs, and appropriate uncertainty, and describe the graph-based context. For the teaching example: “Pearson r was 0.60 (approximate Fisher 95% CI 0.39–0.75; n = 50; $p < 0.001$).” Do not translate this into an original-scale slope.

For linear regression, give the outcome unit and predictor increment. For the constructed example: “Each additional year of age was associated with a 0.50-mmHg higher conditional mean SBP in the simple unadjusted model (95% CI 0.30–0.70; n = 60).” Describe the model and its assessment separately.

For logistic regression, state the event, reference, OR, CI, and adjustment status. Avoid using “independent predictor” when the model contains only one predictor or when the intended meaning of independence has not been justified.

16.9 Build a table around a clear purpose

A table for a mean comparison could contain group, observed n, mean with SD, contrast, and CI. A categorical table could contain counts and percentages with a defined denominator, followed by the chosen measure of association. A model table could contain predictor definition, estimate, interval, and reference categories.

Do not add every software column simply because it is available. Include enough information to assess the result and place secondary output in a supplement when it has a clear role. Define abbreviations in a table note and retain measurement units in the column heading.

An illustrative compact comparison table is:

Quantity	Teaching result
Group A	n = 60; mean 128.0 mmHg; SD 14.0
Group B	n = 60; mean 134.0 mmHg; SD 16.0
Mean difference, A minus B	−6.0 mmHg
95% Welch CI	−11.4 to −0.6 mmHg
Test	t = −2.19; df = 115.96; p = 0.031

The table is explicitly a teaching illustration. Its quantities should not appear under a title implying that they were observed in the author’s institution or patient sample.

16.10 Report figures and uncertainty consistently

Use the same outcome unit, group labels, and contrast direction in the text, table, and figure. Define plotted intervals and any transformation. If a figure shows group means but the text concerns their difference, ensure the relevant contrast is also reported.

For a graph with individual points, identify the observation unit. For a paired graph, describe the pairing and time order. For a model curve, identify whether it shows a fitted mean or an individual prediction interval.

16.11 Precision and rounding

Use precision that reflects the measurement and purpose. Consistent rounding makes a report easier to read, but do not round a value so aggressively that its interpretation changes. A p-value should not be written as 0.000; use an appropriate bound such as $p < 0.001$ or give the small value in scientific notation.

Keep full precision for calculations and round the displayed results at the final reporting step. A CI calculated from rounded intermediate values can differ from one calculated from the original estimate and SE. The accompanying script therefore retains full precision while the manuscript uses readable rounding.

Distinguish a relative percentage change from a percentage-point difference. Increasing a proportion from 15% to 30% is a 15-percentage-point increase and a 100% relative increase. Name the scale to prevent an ambiguous statement.

16.12 Report analyses that did not give a small p-value

A non-significant result still contains an estimate and uncertainty. Report them and explain which values remain compatible with the data and method. The absence of a small p-value does not establish absence of an effect. [36]

Distinguish an inconclusive estimate from a precise estimate near a prespecified negligible range. Equivalence and non-inferiority analyses require their own planned margins and methods. Do not retrofit their conclusions to a conventional comparison.

16.13 Keep the conclusion within the evidence

Use "associated with" for an observational or simple unadjusted relationship unless the design and analysis support a stronger claim. A result from the recruited sample does not automatically generalise to all patients, all facilities, or the entire community.

Separate a statistical estimate from a treatment recommendation. Clinical relevance depends on the clinical objective, benefits and harms, uncertainty, and evidence beyond one analytical result. The invented numerical examples in this book teach interpretation and are not evidence for practice recommendations.

16.14 Preserve an auditable result

Retain the source, cleaned data, dictionary, analysis log, saved jamovi file, software versions, and final exported output. A reviewer should be able to connect the reported result to the analysis settings and the observations used.

Before release, compare the text, table, figure, and output. Check denominators, sign, units, reference categories, intervals, rounding, and p-values. This final consistency check often identifies reporting errors that a successful statistical calculation cannot detect.

16.15 Hands-on

jamovi workspace

16.16 Review questions

1. What is missing from the statement “There was a significant difference”?
2. Why is “OR = 2.43, so risk is 2.43 times higher” incorrect in the teaching example?
3. What should be reported when a comparison is non-significant?
4. Which quantities should agree across the paragraph, table, figure, and saved analysis?

16.17 Answers and reasoning

1. The comparison, direction, magnitude, units, uncertainty, and analytical sample.
2. The OR compares odds, and the example’s risk ratio is 2.00. The scales differ.
3. The estimate, interval, method, denominator, and relevant interpretation of the remaining uncertainty.
4. Denominators, variable definitions, contrast direction, units, reference categories, estimates, intervals, and appropriately rounded inferential results.

Appendix A

Statistical Formulas

This appendix is a reference for the basic quantities used in the chapters. A formula does not establish that its assumptions fit a dataset. Select the estimand and observation structure first, then use the calculation appropriate to them. The cited statistical documentation identifies the corresponding procedures.

A.1 Symbols and direction

Symbol	Meaning
n	Number of independent observations, or complete pairs when specified
$\bar{x}$	Sample mean
s	Sample standard deviation
$\hat{p}$	Sample proportion
$\widehat{\Delta}$	Estimated difference in the defined direction
α	Nominal significance level
$t_{q,\nu}$	Quantile q of a t distribution with degrees of freedom
z_q	Quantile q of the standard normal distribution

For independent comparisons, this appendix uses A minus B or A relative to B. For paired changes, it uses follow-up minus baseline. Define a different direction explicitly if one is used.

A.2 Mean, variance, and standard deviation

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}.$$

The mean and SD retain the measurement unit; variance has the squared unit. The sample SD requires at least two observations. The definitions are described in the NIST location and scale guidance. [21, 23]

The median is the middle ordered value, averaging the two middle observations for an even n. The IQR is $Q_3 - Q_1$. Quantile interpolation conventions can affect small-sample quartiles; the verification script uses R's default type 7. [24]

A.3 Standard error and interval for a mean

For independent observations under the conventional mean model,

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}, \qquad CI_{1-\alpha} = \bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

This is an interval for the population mean, not a range for individual observations. Its coverage depends on the model and procedure. [25]

A.4 Proportion

$$\hat{p} = \frac{\text{number with the defined characteristic}}{\text{appropriate denominator}}, \qquad \text{percentage} = 100\hat{p}.$$

An independent binomial large-sample SE is $\sqrt{\hat{p}(1-\hat{p})/n}$. The corresponding simple Wald interval can perform poorly near zero or one or with limited information. Use a suitable interval procedure and name it; an exact binomial interval is one documented option. [26]

A.5 Independent mean difference: Welch

$$\widehat{\Delta} = \bar{x}_A - \bar{x}_B, \qquad SE(\widehat{\Delta}) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}.$$

The Welch degrees of freedom are

$$\nu = \frac{(s_A^2/n_A + s_B^2/n_B)^2}{\dfrac{(s_A^2/n_A)^2}{n_A - 1} + \dfrac{(s_B^2/n_B)^2}{n_B - 1}}.$$

For a null difference Δ_0,

$$t = \frac{\widehat{\Delta} - \Delta_0}{SE(\widehat{\Delta})}, \qquad CI = \widehat{\Delta} \pm t_{1-\alpha/2,\nu} SE(\widehat{\Delta}).$$

Welch's calculation uses separate variance estimates. A pooled Student method has a different variance assumption and degrees of freedom. [38]

A.6 Independent mean difference: pooled variance

If a shared variance model is appropriate,

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2},$$

$$SE(\widehat{\Delta}) = s_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}, \qquad df = n_A + n_B - 2.$$

Use the pooled SE and its degrees of freedom in the corresponding test and interval. Do not switch between pooled and Welch formulas without identifying the selected model.

A.7 Paired mean change

Let $d_i = x_{i,\text{follow-up}} - x_{i,\text{baseline}}$ for n complete pairs:

$$\bar{d} = \frac{1}{n}\sum_i d_i, \qquad SE(\bar{d}) = \frac{s_d}{\sqrt{n}},$$

$$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}, \quad df = n - 1, \quad CI = \bar{d} \pm t_{1-\alpha/2,n-1}\frac{s_d}{\sqrt{n}}.$$

The relevant spread is the SD of differences, not an independent-group combination of the two measurement SDs. [40]

A.8 Rank quantities

For independent groups with no ties, one U convention is

$$U_A = R_A - \frac{n_A(n_A+1)}{2},$$

where R_A is A's rank sum in the combined sample. With ties, a probability-of-superiority summary gives half credit to tied pairs. The test statistic convention, tie handling, and exact or approximate inference should be identified. [42]

The signed-rank calculation uses ranks of non-zero absolute paired differences, retaining their signs. It has a different estimand and assumptions from the paired mean calculation.

A.9 Conventional one-way ANOVA

For k groups, total N, group means $\bar{x}_j$, and overall mean $\bar{x}$,

$$SS_B = \sum_{j=1}^{k} n_j(\bar{x}_j - \bar{x})^2, \qquad SS_W = \sum_{j=1}^{k}\sum_i (x_{ij} - \bar{x}_j)^2.$$

$$F = \frac{SS_B/(k-1)}{SS_W/(N-k)}, \qquad \eta^2 = \frac{SS_B}{SS_B + SS_W}.$$

These quantities use the conventional independent common-variance model. Welch's one-way procedure uses a different calculation and reference distribution. [43, 44]

A.10 Independent categorical association

$$E_{ij} = \frac{R_i C_j}{N}, \qquad \chi^2 = \sum_{i,j}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad df = (r-1)(c-1).$$

Here R_i and C_j are the row and column totals. Assess the approximation using expected counts and table structure. Yates correction is a 2-by-2 option, not a general sparse-table solution. [49, 50]

For an r-by-c table, a common sample Cramér's V summary based on the uncorrected Pearson statistic is

$$V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}.$$

It is an association summary without a risk or odds direction. Its magnitude should not replace question-specific group contrasts.

A.11 Two-by-two differences and ratios

Use the table orientation:

Group	Event	No event	Total
A	a	b	$n_A = a + b$
B	c	d	$n_B = c + d$

$$p_A = \frac{a}{a+b}, \quad p_B = \frac{c}{c+d}, \quad RD = p_A - p_B, \quad RR = \frac{p_A}{p_B}, \quad OR = \frac{ad}{bc}.$$

A risk interpretation requires an appropriate design and denominator. Odds and probability are different scales. [51]

For independent groups, large-sample formulas are

$$SE(RD) = \sqrt{\frac{p_A(1-p_A)}{n_A} + \frac{p_B(1-p_B)}{n_B}},$$

$$SE(\log RR) = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}},$$

$$SE(\log OR) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}.$$

For the difference, use $RD \pm z_{1-\alpha/2} SE(RD)$. For a ratio, use

$$\exp\{\log(\text{ratio}) \pm z_{1-\alpha/2} SE(\log \text{ratio})\}.$$

The log formulas require positive relevant cell counts. They are approximate intervals, not Fisher exact intervals, and are not a general remedy for sparse or zero-count tables. Appropriate exact, score, or other methods should be considered for those situations.

A.12 Paired binary comparison

If b and c now denote the discordant counts in a paired before–after table, an uncorrected McNemar statistic is

$$\chi^2_M = \frac{(b-c)^2}{b+c}.$$

A continuity-corrected version uses $[\max(|b-c|-1, 0)]^2/(b+c)$. Under the conditional null, the direction count among the discordant pairs can be tested with a binomial probability of one half. If $b + c = 0$, there is no discordant information for the comparison. [26, 52]

A.13 Pearson correlation

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}, \qquad t = r\sqrt{\frac{n-2}{1-r^2}}.$$

The conventional zero-correlation test uses n − 2 degrees of freedom under its model. An approximate Fisher interval forms $z = \operatorname{atanh}(r)$, uses $SE(z) = 1/\sqrt{n-3}$, and back-transforms the limits. Computing r and justifying its inference are separate tasks. [30]

Spearman's coefficient applies Pearson correlation to ranks, with appropriate handling of ties and inference. Do not use a no-ties shortcut indiscriminately when tied ranks are present.

A.14 Simple linear regression

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad e_i = y_i - \hat{y}_i.$$

With an intercept, $R^2 = 1 - SS_E/SS_T$. For one quantitative predictor and the same complete observations, $R^2 = r^2$. Coefficient inference and individual prediction require their own assumptions and uncertainty calculations. [57, 59, 62]

A.15 Simple logistic regression

$$\operatorname{logit}(p_i) = \beta_0 + \beta_1 x_i, \qquad p_i = \frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)}.$$

For a one-unit predictor increase, $OR = \exp(\beta_1)$. A common Wald interval exponentiates $\hat{\beta}_1 \pm z_{1-\alpha/2} SE(\hat{\beta}_1)$. The no-association values are zero on the coefficient scale and one on the OR scale. [61, 63]

Appendix B

Choosing an Appropriate Statistical Method

Use this guide after defining the question, observation unit, and variable meaning. It maps basic questions to methods covered in Volume I and marks situations requiring later-volume extensions. A choice is justified by the estimand and design, not by an attempt to find a small p-value.

B.1 Six decisions before choosing a method

First, identify the population and outcome. Second, define the target quantity: a mean, proportion, contrast, association, or prediction. Third, identify the observation unit. Fourth, determine whether observations are independent, paired, repeated, or clustered. Fifth, inspect coding, missingness, and relevant distributions. Sixth, select the procedure and uncertainty method that match the target and structure.

Document the resulting choice before interpreting the output. If an important assumption is not plausible, reconsider the model or estimand and explain the change.

Use Chapter 6 to reconcile the eligibility rule, required variables, matching, active filters, and analytical sample. The same cleaned file can supply different valid samples for different questions; each result needs its own count and denominator.

B.2 Describe or estimate

Question	Starting method	Required context
How many observations fall in each category?	Frequency table and proportions	Category definitions, denominator, and missingness
What is the central value and spread of a quantitative outcome?	Mean/SD or median/quartile limits suited to the objective	Distribution, units, observed n, and unusual values
How uncertain is a mean estimate?	Appropriate mean CI	Sampling and mean-model assumptions
How uncertain is a proportion estimate?	Appropriate proportion CI, such as a stated exact or score method	Binary definition, denominator, and design

A descriptive summary does not require a hypothesis test solely to describe the sample. Refer to the descriptive chapter for the meaning of location and spread and to the uncertainty chapter for interval interpretation.

B.3 Compare quantitative outcomes

Structure and target	Starting method	Important boundary
Two independent groups; mean difference	Welch or justified pooled independent t procedure	Independence, distributional adequacy, and variance model
Two paired measurements; mean change	Paired t procedure	Complete matching; relevant distribution is that of differences
Two independent groups; ordered/distributional comparison	Mann–Whitney procedure	Rank interpretation; median-shift meaning requires additional conditions
Two paired measurements; signed-rank location question	Wilcoxon signed-rank procedure	Non-zero differences and suitable symmetry for location interpretation
More than two independent groups; mean comparison	Conventional or Welch one-way ANOVA	Variance model and compatible planned/follow-up contrasts
More than two independent groups; rank-based comparison	Kruskal–Wallis with compatible follow-up procedure	Global result does not identify individual pairs
More than two repeated measurements or a clustered outcome	Design-specific repeated/clustered method	Later-volume extension; do not use an ordinary independent-group test

The corresponding statistical documentation distinguishes independent, paired, variance, and rank-based procedures. [38, 42, 44, 46]

B.4 Compare categorical outcomes

Begin with the structure, before expected-count rules:

Structure	Appropriate direction
Independent observations; two categorical variables	Contingency-table association method and design-appropriate contrast
Same individuals with two binary statuses	Paired binary method, such as McNemar or a suitable exact counterpart
Repeated, clustered, matched, or more complex paired categorical data	A procedure that explicitly addresses that structure

For an independent contingency table:

1. Check table dimensions and calculate expected counts under independence.
2. If the approximation is appropriate, a Pearson chi-square procedure may be used.
3. For a sparse 2-by-2 table, consider Fisher's exact procedure or another justified method.
4. For a sparse larger table, consider an appropriate exact or Monte Carlo procedure with documented availability and settings.
5. Select the effect measure and interval using the design, event definition, and reference group.

Yates correction is confined to a 2-by-2 chi-square calculation; it is not the universal destination for sparse tables. Overall sample size cannot replace an expected-count assessment. [49, 50]

For paired binary data, examine discordant pairs and the paired null. A calculation treating before and after observations as independent answers the wrong structure. [52, 53]

B.5 Choose the effect measure

Target	Scale and null value	Interpretation check
Mean or absolute proportion difference	Difference; null zero	Define A minus B and the unit or percentage points
Risk ratio	Ratio; null one	Follow-up and denominators support a risk interpretation
Prevalence ratio	Ratio; null one	Cross-sectional target and denominator are explicit
Odds ratio	Ratio; null one	Compare odds; do not call it risk
Correlation	Coefficient; zero-linear or zero-rank association as specified	Define coefficient, pairs, relationship form, and interval method
Regression slope	Outcome units per predictor increment; null zero	Name outcome, predictor, unit, and adjustment status

An available software measure can be inappropriate to the sampled design. In outcome-based sampling, the observed case proportions are not directly population risks. The odds-ratio chapter explains the difference between odds and probabilities. [51]

B.6 Association and simple modeling

Question	Starting method	Scope limit
Linear association between two quantitative variables	Pearson correlation with scatterplot and appropriate inference	Conventional independent-pair model; not causation or agreement
Monotonic ordered association	Spearman correlation	Rank-based target; a U-shaped relationship is not generally monotonic
Continuous outcome with one predictor	Simple linear regression	Assess mean form and residual assumptions; unadjusted
Binary outcome with one predictor	Introductory logistic model	Defined event, logit scale, and basic model checks; unadjusted
Adjusted effect, count outcome, survival, repeated data, or diagnostic accuracy	Appropriate specialist model	Volume II
Causal effect identification, complex strategy, predictive validation, or evidence synthesis	Explicit advanced framework	Volume III

The software and statistical references identify the implemented coefficients and models. Their availability does not define the scientific objective. [30, 57, 61]

B.7 When assumptions are doubtful

A doubtful assumption does not automatically identify one replacement test. Determine what failed and whether the intended quantity remains meaningful. Unequal variances, a curved mean relationship, paired observations, and a highly skewed distribution are different problems requiring different responses.

For example, replacing Pearson with Spearman can address a monotonic rank-based objective but does not summarise an arbitrary curve. Replacing a pooled t-test with Welch relaxes equal variance but does not handle clustering. Replacing a paired mean test with signed ranks changes the target and requires its own assumptions.

B.8 A final selection statement

Write the choice as a sentence: "The observations are independent, the outcome is continuous, and the primary estimand is the mean in A minus B; therefore, the analysis uses a specified independent mean-difference method with a 95% CI." For a paired study, the sentence should identify the within-pair change instead.

Then read the relevant chapter. Confirm the data, checks, selected options, output meaning, and reporting. This statement makes a method choice reviewable and helps distinguish a justified analysis from a software-driven choice.

Appendix C

Glossary

These definitions use the meanings applied in Volume I. Follow the relevant chapter when a quantity has several possible conventions or requires design-specific interpretation.

C.1 Data and design

Term	Meaning in this book
Analysis-ready data	A reviewed file with documented coding, eligibility, units, and missing-value rules
Analysis specification	A stated question, target quantity, eligible sample, variables, method, and reporting plan
Binary variable	A variable with two defined categories
Categorical variable	A variable whose values indicate categories rather than a continuous numerical amount
Complete case	An observation with values for all variables required by the specified analysis
Complete pair	Two matched measurements both available for the paired calculation
Continuous variable	A quantitative measurement treated on a numerical scale with meaningful distances
Data dictionary	Definitions of variables, units, categories, timing, missing codes, and derivations
Data type	Storage property such as integer, decimal, or text
Data validation	Checks of whether the file follows its measurement and entry definitions
Derived variable	A new quantity calculated or recoded from documented source variables
Filter	A defined rule controlling which observations enter analyses
Identifier	A label used to identify or link observation units
Independent observations	Observations treated as independent under the design and analysis model; repeated or clustered records need assessment
Long format	Data with measurements across occasions represented in separate rows and linked by identifiers
Measure type	Analytical property such as nominal, ordinal, continuous, or ID
Missing value	An unavailable intended measurement or status; not automatically a zero or negative result
Missing at random (MAR)	Missingness independent of missing values conditional on relevant observed information
Missing completely at random (MCAR)	Missingness unrelated to observed and unobserved values
Missing not at random (MNAR)	Dependence of missingness on unobserved values remains
Nominal variable	Categories with no inherent order
Ordinal variable	Ordered categories without an automatic equal-spacing interpretation
Paired data	Observations related by a defined match, such as two measurements from the same participant

Term	Meaning in this book
Row key	A variable or combination of variables identifying a row under the defined observation unit
Unit of observation	The entity represented by an analytical record, such as a patient, visit, admission, or facility
Wide format	Data with different measurements represented in separate columns for the same unit

The software distinctions between data type, measure type, and derived variables are documented in jamovi. Complete-case status is defined relative to the variables used in a calculation. Missingness assumptions concern the data process. [4, 11, 12, 14]

C.2 Description and uncertainty

Term	Meaning in this book
Confidence interval	An interval produced by a procedure with a stated coverage rate under its assumptions
Denominator	The set or count relative to which a proportion is calculated
Estimate	A sample-based value used to learn about a target parameter
Estimand	The precisely defined quantity the analysis aims to estimate
Frequency	The number of observations in a category or interval
Geometric mean	For positive values, the exponential of the average of their natural logarithms
Interquartile range	Third quartile minus first quartile
Mean	Sum of the observations divided by their number
Median	Middle ordered value, with the two middle values averaged when n is even
Mode	Most frequent value or category; it need not be unique
Parameter	A quantity describing the population or statistical model
Percentage point	A unit for an absolute difference between percentages
Precision	The degree of uncertainty in an estimate under the selected procedure
Proportion	A defined count divided by an appropriate denominator
Quartile	A value dividing an ordered distribution at a specified quarter; interpolation conventions can vary
Range	The span from minimum to maximum, or their difference when explicitly stated
Sampling variability	Variation in a statistic across repeated samples under the sampling model
Standard deviation	Dispersion of observations around their mean, in the original unit
Standard error	Estimated sampling variability of an estimator, distinct from individual spread
Standardisation	Expressing a value relative to a stated reference mean and SD
Statistic	A quantity calculated from the observed sample
Variance	Squared standard deviation under the corresponding definition

The location, spread, quantile, and mean-interval definitions are described in the statistical reference documentation. [21, 23–25]

C.3 Testing and interpretation

Term	Meaning in this book
Alternative hypothesis	The specified departure from the null evaluated by a test
Bias	A systematic problem in an estimator or study process relative to its target; not ordinary sampling variation

Term	Meaning in this book
Clinical importance	Relevance of an effect magnitude to the stated clinical objective, distinct from a p-value threshold
Contrast	A defined comparison, such as A minus B or A relative to B
Degrees of freedom	A parameter of a test's reference distribution, determined by its procedure
Multiplicity	The problem of interpreting several related analyses and their combined error criterion
Null hypothesis	The specified value or relationship evaluated under a test model
Omnibus test	An overall assessment across a set of groups or terms rather than one specific pair
P-value	Probability of the chosen test statistic being at least as incompatible as observed, conditional on the specified null model
Post-hoc comparison	A follow-up comparison whose procedure and multiplicity handling must be identified
Power	Probability of rejecting the null under a specified alternative and test procedure
Reference category	The category against which a contrast or model coefficient is expressed
Significance level	The nominal Type I error control level under the test's assumptions
Type I error	Rejecting a true null under the test procedure
Type II error	Failing to reject the null under a specified true alternative

Interpretation of p-values, errors, and power depends on the conditioning and model. A non-significant result does not alone establish equivalence. [1, 34–36]

C.4 Association and modeling

Term	Meaning in this book
Cramér's V	A scaled categorical-association summary based on a chi-square statistic and table dimensions
Expected count	Cell frequency implied by the specified table model, such as independence
Fitted value	The model's estimated response quantity at a specified predictor value
Homoscedasticity	A constant error variance under the relevant model
Influence	The extent to which an observation affects the fitted result
Intercept	The model quantity at the predictor's reference or zero value
Leverage	A property related to an observation's predictor position relative to the fitted design
Linear association	A relationship summarised by a straight-line pattern
Logit	Natural logarithm of probability divided by its complement
Monotonic association	An ordered relationship that moves in one direction, not necessarily in a straight line
Odds	Probability divided by one minus that probability
Odds ratio	Ratio of two odds; the null value is one
Outcome	The variable whose distribution or relationship is the focus of the analysis
Pearson correlation	A standardised summary of linear association between paired quantitative values
Prediction interval	An interval for a new observation under a model, including individual variation
Predictor	A variable used to specify or explain a model outcome; the role does not establish causation
Residual	Observed value minus fitted value for the relevant model quantity
Risk difference	Difference between design-supported risks, often expressed in percentage points
Risk ratio	Ratio of design-supported risks
R-squared	In ordinary linear regression with an intercept, the share of observed squared outcome variation accounted for by the fit
Separation	A binary-model data pattern that prevents a finite ordinary maximum-likelihood estimate in the affected direction

Term	Meaning in this book
Slope	Change in the modeled outcome quantity for a defined predictor increment
Spearman correlation	Correlation of ranks used to summarise monotonic association
Unadjusted model	A model that does not control for additional characteristics beyond its stated predictor structure

The categorical, correlation, linear, and generalised-linear-model references provide the associated calculations and output definitions. [30, 49, 51, 57, 59, 61, 62]

Appendix D

Practice Questions and Applied Cases

This appendix contains conceptual and numerical exercises that can be completed from the teaching text. All values are invented. Downloadable jamovi datasets and screenshots are reserved for the Hands-on sections and will be supplied by the author.

D.1 Case 1: Prepare the analysis question

A clinic file contains participant ID, group A/B, baseline SBP, follow-up SBP, and event status. Some participants have more than one visit record, and event status is unavailable for some.

D.1.1 Tasks

1. Define an independent-group mean question and a paired-change question.
2. Identify the observation unit required for each simple analysis.
3. Explain why an unknown event status should not be coded as absence.
4. Specify the information needed to select one follow-up record per participant.

D.1.2 Reasoning and answers

An independent question can ask for the mean follow-up SBP in A minus B among eligible participants with a defined follow-up measurement. A paired question can ask for mean follow-up minus baseline SBP among complete pairs. The first requires one outcome per participant in one group; the second requires matching the two measurements within each participant.

Unknown status does not establish no event. Record missingness and ascertainment separately. To select one follow-up record, define the time window and selection rule from the protocol, such as proximity to the planned visit. A choice based on which result is favourable is not a valid record-selection rule.

D.2 Case 2: Summarise waiting times

The invented values are 2, 3, 3, 4, 4, 5, 6, 7, 8, and 18 minutes.

D.2.1 Tasks

1. Calculate the mean and median.
2. Calculate the sample SD and first–third quartiles using the script's type-7 convention.

3. Explain what the value of 18 does to the mean relative to the median.
4. Select a graph and write a two-sentence descriptive result.

D.2.2 Reasoning and answers

The mean is 6.0 minutes, median 4.5, sample SD 4.62, and first–third quartiles 3.25–6.75. The IQR is 3.50 and observed limits are 2–18 minutes. The value of 18 extends the upper tail and increases the mean relative to the central ordered values.

A histogram with a boxplot or individual points can show concentration and the long upper tail. A result could read: "Waiting time was available for 10 observations. The median was 4.5 minutes (first–third quartile 3.25–6.75), and values ranged from 2 to 18 minutes; the mean was 6.0 minutes (SD 4.6)."

The separate value is not automatically an error. Verify its source before deciding whether any analytical treatment is needed.

D.3 Case 3: Compare independent groups

Group A has n = 60, mean SBP 128 mmHg, and SD 14. Group B has n = 60, mean 134, and SD 16. Assume the independent structure and conventional Welch inference are suitable for the illustration.

D.3.1 Tasks

1. Define the contrast A minus B and calculate its estimate.
2. Reproduce the Welch SE, degrees of freedom, and 95% interval.
3. Interpret the result without relying on the p-value threshold alone.
4. Assess whether the interval establishes a reduction of at least an illustrative 5 mmHg.

D.3.2 Reasoning and answers

The estimate is −6.0 mmHg, SE 2.745, degrees of freedom approximately 115.96, and 95% CI −11.44 to −0.56. The two-sided p-value is 0.0308.

The estimate is negative in the stated direction and the interval excludes zero under the model. However, the interval includes reductions smaller than 5 mmHg as well as larger reductions. It therefore does not establish a reduction of at least that illustrative size. Whether the difference can be attributed to treatment depends on design and conduct not specified by these summaries.

D.4 Case 4: Compare paired measurements

There are 30 complete pairs. Mean follow-up minus baseline SBP is −8 mmHg and the SD of differences is 10 mmHg.

D.4.1 Tasks

1. Calculate the paired SE and t statistic.
2. State the degrees of freedom and 95% CI.
3. Identify the distribution that needs assessment for the conventional paired t inference.
4. Explain why the result does not alone establish the cause of the change.

D.4.2 Reasoning and answers

The SE is 1.826 mmHg, t = −4.382, df = 29, and CI −11.73 to −4.27. The two-sided p-value is 0.000141. The relevant small-sample distributional assessment concerns the within-pair differences.

Matching and dependence between pairs also need to be checked. A before–after change can be associated with several time-related processes, so a non-zero mean change does not itself isolate an intervention effect.

D.5 Case 5: Interpret a three-group analysis

Three groups each have 20 observations and SD 10 mmHg; their means are 125, 130, and 135 mmHg. The conventional one-way illustration gives $F(2, 57) = 5.00$ and $p = 0.00998$.

D.5.1 Tasks

1. State the null and alternative hypotheses.
2. Explain why the omnibus result does not identify all pairwise differences.
3. Identify what must be specified for follow-up comparisons.
4. Explain why three repeated time points would require a different analysis structure.

D.5.2 Reasoning and answers

The null states that all population means are equal. The alternative states that they are not all equal. At least one departure can produce the omnibus result; it does not establish that each pair differs.

Specify the contrasts or pairs, the variance model, compatible follow-up procedure, and multiplicity family. Measurements repeated in the same people are dependent and cannot be treated as three independent samples by this model.

D.6 Case 6: Compare event proportions

Assume an invented cohort-like comparison with complete common follow-up. A has 30 events and 70 non-events; B has 15 events and 85 non-events.

D.6.1 Tasks

1. Calculate group-specific proportions and the absolute difference.
2. Calculate RR and OR for A relative to B.
3. Calculate expected counts and identify the Pearson test result.
4. Explain why OR and RR should not be interchanged.
5. Explain how the interpretation would change in outcome-based sampling.

D.6.2 Reasoning and answers

The proportions are 30% and 15%, the difference is 15 percentage points, RR = 2.00, and OR = 2.43. Expected counts are 22.5 and 77.5 in each row. The uncorrected Pearson statistic is 6.452 with df = 1 and $p = 0.0111$.

RR compares probabilities; OR compares odds. The log-Wald intervals are 1.15–3.48 for RR and 1.21–4.87 for OR. Fisher's two-sided $p = 0.0171$ is a different inferential calculation, not an alternative selected because it is preferable numerically.

If participants were selected as cases and controls, the sampled event fractions would not directly estimate the population risks. The design must support the interpretation of the displayed denominator and effect measure.

D.7 Case 7: Preserve categorical pairing

In 100 invented before–after pairs, 20 are present at both times, 15 change present to absent, 5 change absent to present, and 60 remain absent.

D.7.1 Tasks

1. Calculate the before and after proportions.
2. Identify the discordant pairs.
3. Name the structure-appropriate comparison and explain why independent testing is unsuitable.

D.7.2 Reasoning and answers

The before proportion is 35% and after proportion 25%. The discordant counts are 15 and 5, totalling 20. A McNemar-type or suitable exact paired comparison retains the matching. Treating before and after as independent samples would discard it.

The exact conditional two-sided binomial p-value for 15 of 20 discordant pairs in one direction under probability 0.5 is 0.0414. Its interpretation and interval require a specified paired procedure; this teaching result should not be mistaken for the independent-table example.

D.8 Case 8: Correlation and regression interpretation

Consider two separate invented summaries: Pearson r = 0.60 from 50 pairs, and a simple linear slope of 0.50 mmHg per year with SE 0.10 from 60 observations.

D.8.1 Tasks

1. Interpret the coefficient of correlation and its Fisher interval.
2. Explain why the correlation does not give a slope in mmHg/year.
3. Interpret the slope and its approximate 95% interval.
4. Explain why a perfectly correlated pair of measurement methods can disagree.

D.8.2 Reasoning and answers

The correlation describes positive linear association, with an approximate Fisher CI of 0.39–0.75. It is unitless and does not quantify the original-scale increment. The slope indicates a 0.50-mmHg difference in conditional mean for each year in the simple unadjusted model, with CI approximately 0.30–0.70.

If method B equals A plus 20 units, their Pearson correlation can be +1 while a systematic difference remains. Agreement and correlation therefore answer different questions.

D.9 Case 9: Translate log odds

In the simple model for the independent event table, group B is coded 0, A is coded 1, and the group coefficient is 0.8873.

D.9.1 Tasks

1. Exponentiate the coefficient.
2. State the no-association value on each scale.
3. Explain what is and is not adjusted by this model.
4. Write one result sentence using the correct outcome scale.

D.9.2 Reasoning and answers

The exponentiated coefficient is approximately 2.43. The null is zero for log odds and one for the OR. The model contains group only and has not adjusted for other characteristics.

A sentence could read: "Group A had higher event odds than B in the simple unadjusted model (OR 2.43; Wald 95% CI 1.21–4.87)." It should not replace odds with risk or claim independent adjustment.

D.10 Case 10: Audit a results paragraph

Consider the deliberately flawed statement: "The treatment significantly improved the outcome ($p = 0.000$), and there was no difference in the other outcome ($p = 0.08$)."

D.10.1 Tasks

Identify the missing information and unsupported conclusions. Rewrite the structure using placeholders for information that would be required from the actual study.

D.10.2 Reasoning and answers

The statement omits the outcome definition, analytical sample, groups, effect magnitude, units, intervals, test, and design. "Improved" asserts a favourable and potentially causal interpretation without the required context. $p = 0.000$ is an inappropriate rounding statement. $p = 0.08$ does not establish no difference.

A defensible structure is: "Among [analytical n], [outcome] differed between [defined groups] by [estimate and unit] ([interval and method]; [test and p-value]). For [second outcome], the estimate was [value] with CI [limits], indicating [what the remaining uncertainty permits]." Complete the placeholders from the actual analysis rather than inventing the missing study results.

D.11 Practical datasets to be added by the author

The software exercises will need datasets with documented variables and the appropriate structure:

Dataset purpose	Required structure	Core variable definitions
Description and independent comparisons	One eligible participant per row	ID, group, outcome, unit, and missingness
Paired comparison	Matched baseline and follow-up values	ID, timing, paired measurements, and difference direction
Categorical association	Independent group and known event status	Group, event definition, follow-up, and reference categories
Correlation and simple modeling	Complete predictor–outcome pairs	ID, quantitative variables, units, and included range

Provide a data dictionary, source or generation description, and version information with each file. The numerical illustrations above remain separate teaching calculations unless a practical dataset is explicitly designed and checked to reproduce them.

D.12 Hands-on

jamovi workspace

References

The sources below support the statistical definitions, analysis assumptions, interpretation, and software features cited in the text. Software documentation was checked on 5 October 2026. Menu labels and available options may differ by jamovi or module version; the Hands-on demonstrations should record the versions used.

1. American Statistical Association. Statement on statistical significance and p-values. https://www.amstat.org/asa/files/pdfs/p-valuestatement.pdf [accessed 5 Oct. 2026].
2. EQUATOR Network. Basic statistical reporting for articles published in biomedical journals: The SAMPL guidelines. https://www.equator-network.org/reporting-guidelines/sampl/ [accessed 5 Oct. 2026].
3. EQUATOR Network. STROBE: Guidelines for reporting observational studies. https://www.equator-network.org/reporting-guidelines/strobe/ [accessed 5 Oct. 2026].
4. jamovi. Data variables. https://docs.jamovi.org/data/data_1_overview_data_variables.html [accessed 5 Oct. 2026].
5. Pennsylvania State University. What is data: Measurement scales. https://online.stat.psu.edu/stat857/node/31/ [accessed 5 Oct. 2026].
6. Greenland S, Senn SJ, Rothman KJ, Carlin JB, Poole C, Goodman SN, et al. Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. European Journal of Epidemiology. 2016;31:337–350. doi: 10.1007/s10654-016-0149-3 https://pmc.ncbi.nlm.nih.gov/articles/PMC4877414/
7. jamovi. Getting started. https://www.jamovi.org/getting-started.html [accessed 5 Oct. 2026].
8. jamovi. The spreadsheet. https://docs.jamovi.org/usermanual/um_4_spreadsheet.html [accessed 5 Oct. 2026].
9. jamovi. Combining jamovi and r. https://docs.jamovi.org/usermanual/um_6_jamovi_and_R.html [accessed 5 Oct. 2026].
10. Broman KW, Woo KH. Data organization in spreadsheets. The American Statistician. 2018;72(1):2–10. doi: 10.1080/00031305.2017.1375989 https://kbroman.org/Paper_DataOrg/
11. Rubin DB. Inference and missing data. Biometrika. 1976;63(3):581–592. doi: 10.1093/biomet/63.3.581 https://doi.org/10.1093/biomet/63.3.581
12. R Core Team. Find complete cases. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/complete.cases.html [accessed 5 Oct. 2026].
13. jamovi. One-way ANOVA: anovaOneW. https://docs.jamovi.org/jmv/jmv_anovaOneW.html [accessed 5 Oct. 2026].
14. jamovi. Computed variables. https://docs.jamovi.org/data/data_2_computed_variables.html [accessed 5 Oct. 2026].
15. jamovi. Transformed variables. https://docs.jamovi.org/data/data_3_transformed_variables.html [accessed 5 Oct. 2026].
16. jamovi. List of functions. https://docs.jamovi.org/data/data_5_list_of_functions.html [accessed 5 Oct. 2026].

17. World Health Organization. Malnutrition in women: Body mass index definition. https://www.who.int/data/nutrition/nlis/info/malnutrition-in-women [accessed 5 Oct. 2026].
18. Altman DG, Royston P. The cost of dichotomising continuous variables. BMJ. 2006;332(7549):1080. doi: 10.1136/bmj.332.7549.1080 https://www.bmj.com/content/332/7549/1080.1
19. jamovi. Introducing filters in jamovi. https://blog.jamovi.org/2018/04/25/jamovi-filters.html [accessed 5 Oct. 2026].
20. NIST/SEMATECH. Histogram. https://www.itl.nist.gov/div898/handbook/eda/section3/eda33e.htm [accessed 5 Oct. 2026].
21. NIST/SEMATECH. Measures of scale. https://www.itl.nist.gov/div898/handbook/eda/section3/eda356.htm [accessed 5 Oct. 2026].
22. jamovi. Contingency tables: contTables. https://docs.jamovi.org/jmv/jmv_contTables.html [accessed 5 Oct. 2026].
23. NIST/SEMATECH. Measures of location. https://www.itl.nist.gov/div898/handbook/eda/section3/eda351.htm [accessed 5 Oct. 2026].
24. R Core Team. Sample quantiles. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/quantile.html [accessed 5 Oct. 2026].
25. NIST/SEMATECH. Confidence limits for the mean. https://www.itl.nist.gov/div898/handbook/eda/section3/eda352.htm [accessed 5 Oct. 2026].
26. R Core Team. Exact binomial test. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/binom.test.html [accessed 5 Oct. 2026].
27. jamovi. Descriptives: descriptives. https://docs.jamovi.org/jmv/jmv_descriptives.html [accessed 5 Oct. 2026].
28. NIST/SEMATECH. Box plot. https://www.itl.nist.gov/div898/handbook/eda/section3/boxplot.htm [accessed 5 Oct. 2026].
29. NIST/SEMATECH. Normal probability plot. https://www.itl.nist.gov/div898/handbook/eda/section3/normprpl.htm [accessed 5 Oct. 2026].
30. R Core Team. Test for association/correlation between paired samples. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/cor.test.html [accessed 5 Oct. 2026].
31. International Committee of Medical Journal Editors. Preparing a manuscript for submission to a medical journal. https://www.icmje.org/recommendations/browse/manuscript-preparation/preparing-for-submission.html [accessed 5 Oct. 2026].
32. NIST/SEMATECH. Normal distribution. https://www.itl.nist.gov/div898/handbook/eda/section3/eda3661.htm [accessed 5 Oct. 2026].
33. Pennsylvania State University. The central limit theorem: STAT 414. https://online.stat.psu.edu/stat414/Lesson27 [accessed 5 Oct. 2026].
34. NIST/SEMATECH. Quantitative techniques: Hypothesis tests. https://www.itl.nist.gov/div898/handbook/eda/section3/eda35.htm [accessed 5 Oct. 2026].
35. R Core Team. Power calculations for one and two sample t tests. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/power.t.test.html [accessed 5 Oct. 2026].
36. Altman DG, Bland JM. Absence of evidence is not evidence of absence. BMJ. 1995;311:485. doi: 10.1136/bmj.311.7003.485 https://www.bmj.com/content/311/7003/485
37. R Core Team. Adjust p-values for multiple comparisons. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/p.adjust.html [accessed 5 Oct. 2026].
38. R Core Team. Student's t-test. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/t.test.html [accessed 5 Oct. 2026].
39. jamovi. T-test: Independent samples. https://docs.jamovi.org/analyses/jg__21__t-test-independent.html [accessed 5 Oct. 2026].

40. NIST/SEMATECH. Analysis of paired observations. https://www.itl.nist.gov/div898/handbook/prc/section3/prc311.htm [accessed 5 Oct. 2026].
41. jamovi. Paired samples t-test: ttestPS. https://docs.jamovi.org/jmv/jmv_ttestPS.html [accessed 5 Oct. 2026].
42. R Core Team. Wilcoxon rank sum and signed rank tests. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/wilcox.test.html [accessed 5 Oct. 2026].
43. NIST/SEMATECH. The one-way ANOVA model and assumptions. https://www.itl.nist.gov/div898/handbook/prc/section4/prc432.htm [accessed 5 Oct. 2026].
44. R Core Team. Test for equal means in a one-way layout. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/oneway.test.html [accessed 5 Oct. 2026].
45. R Core Team. Compute tukey honest significant differences. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/TukeyHSD.html [accessed 5 Oct. 2026].
46. R Core Team. Kruskal-wallis rank sum test. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/kruskal.test.html [accessed 5 Oct. 2026].
47. R Core Team. Pairwise wilcoxon rank sum tests. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/pairwise.wilcox.test.html [accessed 5 Oct. 2026].
48. jamovi. One-way ANOVA, non-parametric: anovaNP. https://docs.jamovi.org/jmv/jmv_anovaNP.html [accessed 5 Oct. 2026].
49. R Core Team. Pearson's chi-squared test for count data. https://www.stat.ethz.ch/R-manual/R-patched/library/stats/html/chisq.test.html [accessed 5 Oct. 2026].
50. R Core Team. Fisher's exact test for count data. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/fisher.test.html [accessed 5 Oct. 2026].
51. Bland JM, Altman DG. The odds ratio. BMJ. 2000;320:1468. doi: 10.1136/bmj.320.7247.1468 https://www.bmj.com/content/320/7247/1468.1
52. R Core Team. McNemar's chi-squared test for count data. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/mcnemar.test.html [accessed 5 Oct. 2026].
53. jamovi. Paired samples contingency tables: contTablesPaired. https://docs.jamovi.org/jmv/jmv_contTablesPaired.html [accessed 5 Oct. 2026].
54. jamovi. Correlation matrix: corrMatrix. https://docs.jamovi.org/jmv/jmv_corrMatrix.html [accessed 5 Oct. 2026].
55. jamovi. Linear regression: linReg. https://docs.jamovi.org/jmv/jmv_linReg.html [accessed 5 Oct. 2026].
56. jamovi. From SPSS to jamovi: Logistic regression. https://docs.jamovi.org/spss2jamovi/s2j_logReg.html [accessed 5 Oct. 2026].
57. R Core Team. Fitting linear models. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/lm.html [accessed 5 Oct. 2026].
58. NIST/SEMATECH. How can i tell if a model fits my data? https://www.itl.nist.gov/div898/handbook/pmd/section4/pmd44.htm [accessed 5 Oct. 2026].
59. R Core Team. Predict method for linear model fits. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/predict.lm.html [accessed 5 Oct. 2026].
60. R Core Team. Family objects for models. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/family.html [accessed 5 Oct. 2026].
61. R Core Team. Fitting generalized linear models. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/glm.html [accessed 5 Oct. 2026].
62. R Core Team. Summarizing linear model fits. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/summary.lm.html [accessed 5 Oct. 2026].
63. R Core Team. Summarizing generalized linear model fits. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/summary.glm.html [accessed 5 Oct. 2026].